A

PROJECT REPORT

ON

" MAHARASHTRA BUS PASS SYSTEM "

SUBMITTED TO

SHIVAJI UNIVERSITY, KOLHAPUR

FOR THE PARTIAL FULLFILLMENT OF DEGREE

OF

BACHELOR OF COMPUTER APPLICATIONS

BCA – III

SEM- VI (2025-2026)

UNDER FACULTY OF COMMERCE

BY

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Certificate

This is to certify that the project entitled **" MAHARASHTRA BUS PASS SYSTEM "** which is being submitted here with for the partial fulfillment of the degree of Bachelor of Computer Applications as laid down by Shivaji University, Kolhapur, is the original work completed in this college by **Rohit Malatwadkar & Nikhil Kalekar** under my supervision and guidance and to the best of my knowledge has not submitted to any other University or examining body**.**

Place: Kolhapur

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(H.O.D of B.C.A)

Mr K.V. Kaingade **Examiner**

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DEPARTMENT OF COMPUTER APPLICATIONS

Guide’s Certificate

This is to certify that **Rohit Malatwadkar & Nikhil Kalekar of (B.C.A III)** has completed him project report on the topic**, “ MAHARASHTRA BUS PASS SYSTEM ”** in Python, Flask. For the partial fulfillment of the B.C.A.III submitted to the **“ Shivaji University, Kolhapur** ” during the academic year **2025-26** under my guidance.

To the best of my knowledge & belief the matter presented by him is original and have not been copied from any source**.**

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Gopal Krishna Gokhale

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DEPARTMENT OF COMPUTER APPLICATIONS

Declaration

We hereby declare that the project entitled **" MAHARASHTRA BUS PASS SYSTEM** " have been developed and completed by us are our original work and have not previously been submitted to any University or examining body for the award of any degree.

We have referred the books given in the bibliography during the development of the project and have not copied any of written material or its part thereof. This project is purely on our own and on our merits.

Place: Kolhapur

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Student Name

Rohit Ranaba Malatwadkar

Nikhil Tanaji Kalekar

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Last but not least, we would like to express our sincere gratitude to those who have helped me directly and indirectly in our project.

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Chapter 1:

Introduction To The Project

1.1 – INTRODUCTION

The Maharashtra Bus Pass System is a comprehensive digital solution designed to streamline the process of issuing, managing, and validating bus passes for students across the state of Maharashtra.

- This system aims to replace the traditional manual method of issuing bus passes, which often involves lengthy queues, paperwork, and administrative inefficiencies. The system provides a centralized platform where students from schools, colleges, and other educational institutions can apply for bus passes online, track their application status, and receive digital or physical passes after verification.

  The system is designed to integrate with the Maharashtra State Road Transport Corporation (MSRTC), which is the primary public transport authority in the state. By digitizing the entire process, the system ensures transparency, reduces processing time, and eliminates the possibility of fraudulent passes.

- The platform is built to handle a large volume of applications efficiently, ensuring that students can access affordable transportation without unnecessary delays. Additionally, the system maintains a complete database of all issued passes, allowing authorities to monitor usage patterns and make data-driven decisions for improving public transport services for students.

  The main aim of this system is to reduce paperwork and improve the efficiency of the bus pass management process. At the same time, it gives administrators a centralized dashboard where they can manage all applications, approve or reject requests, and maintain records digitally.

- This project is developed using basic web technologies such as HTML and CSS for designing the frontend, and Python Flask for handling backend operations. The database is managed using SQL, which ensures proper storage and retrieval of user data.

  Another important feature of this system is the application status tracking. After submitting the form, users can check whether their bus pass is approved, pending, or rejected. This adds transparency to the process and reduces the need for repeated visits to the office.

1.2 WORKING OF SYSTEM

The working of the Maharashtra Bus Pass System follows a structured workflow that begins with student registration and ends with pass issuance and validation.

- Students first create an account on the system by providing their basic personal details, educational institution information, and contact information. Once registered, they can log in to fill out the bus pass application form, which requires details such as route information, boarding and dropping points, type of pass (monthly, quarterly, half-yearly, or annual), and uploading necessary documents including student ID card, institution bonafide certificate, address proof, and photograph.

  After submission, the application is forwarded to the respective educational institution for verification. The institution verifies the student's enrollment status and confirms that the student is genuinely pursuing education. Upon institutional approval, the application moves to the concerned bus depot or regional transport office under MSRTC for final approval.

- The transport authority verifies the route availability, fare calculation, and eligibility criteria. Once approved, the student receives a notification and can download the digital bus pass or collect a physical pass from the designated counter.

  The system also incorporates a validation mechanism where bus conductors can verify the authenticity of the pass using a unique QR code or pass number. The system maintains logs of all activities, ensuring accountability and transparency throughout the process. Overall, the Online Bus Pass Management System is a practical solution that demonstrates how technology can be used to improve traditional systems.

- It not only saves time and effort but also ensures better data management and accessibility. Renewals and pass modifications are also handled through the same system, allowing students to apply for pass extensions without visiting the office repeatedly.

  This project reflects the importance of digital transformation in public services and can be further enhanced in the future by adding features like online payment, QR-based passes, and mobile application support.

1.3 NEED OF THE SYSTEM

In the current scenario, the bus pass system in many regions of India is still managed manually. Students and daily commuters are required to visit bus depots or offices physically to apply for a bus pass. This process involves filling out paper forms, submitting documents, and waiting in long queues for approval. Such a system is not only time-consuming but also inefficient and inconvenient for users.

- One of the major issues with the manual system is the lack of proper record management. Since most of the data is handled on paper, there is a high possibility of data loss, misplacement of documents, and errors in maintaining records. It also becomes difficult for the authorities to track applications and verify information quickly.

  Another important problem is the delay in the approval process. In the traditional system, the application needs to be verified manually by the staff, which can take several days. Students often have to visit the office multiple times just to check the status of their application. This leads to unnecessary effort, wastage of time, and frustration among users.

- The manual system also lacks transparency. Applicants are not able to track the status of their bus pass easily, and there is no clear communication regarding approval or rejection. This creates confusion and reduces trust in the system.

  To overcome all these issues, there is a strong need for a digital and automated solution. The Online Bus Pass System fulfills this need by providing a simple and efficient platform where users can apply for bus passes from anywhere, without the need to visit any office physically. It reduces paperwork and minimizes human errors by storing all the data in a structured database.

- In addition, the system provides a feature for checking application status, which increases transparency and keeps users informed. This reduces the need for repeated visits and makes the entire process more user-friendly.

  Therefore, the development of an Online Bus Pass System is essential to modernize the existing system, improve efficiency, save time, and provide a better experience for both users and administrators.

1.4 SCOPE OF THE SYSTEM

The scope of the Online Bus Pass System is to provide a simple, efficient, and digital platform for managing the process of bus pass applications and approvals. This system is mainly designed for students and administrators, where students can apply for a bus pass online and administrators can verify and manage the applications through a centralized system.

- The system covers the basic functionalities required for a bus pass management process. It allows users to fill out an application form by entering necessary details such as name, college, travel route, and type of pass. Once the form is submitted, the data is stored in a database and can be accessed by the administrator for further processing.

- On the administrative side, the system provides features to view all submitted applications, approve or reject them, and update the status accordingly. This ensures that the entire process is handled digitally without the need for paperwork. The system also allows users to check the status of their application, which adds transparency and improves user experience.

- The scope of this project is currently limited to a basic web-based application that demonstrates the core concept of digital bus pass management. It does not include advanced features such as online payment, document verification using AI, or integration with government transport systems. However, these features can be added in the future to enhance the functionality of the system.

- This system can be implemented at a small level, such as within a college or a local transport department, and can later be expanded to a larger scale. In the future, the system can be upgraded to support mobile applications, QR code- based digital passes, SMS or email notifications, and real-time tracking of bus routes.

  Overall, the scope of the Online Bus Pass System is to replace the manual process with a digital solution that is easy to use, reliable, and scalable for future enhancements.

Chapter 2: Proposed System

2.1 INTRODUCTION TO PROPOSED SYSTEM

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The proposed system, titled "Maharashtra Bus Pass System," is designed as a digital platform to manage the bus pass application and approval process for students who use state transport buses for educational purposes. This system mainly focuses on students from schools, colleges, and other educational institutions within Maharashtra who depend on public bus services for their daily travel.

- In the existing system, the bus pass process is handled manually, which involves physical form submission, long queues, and delays in approval. To overcome these challenges, the proposed system introduces an online solution where students can apply for bus passes easily through a web-based interface. The system allows users to enter their personal and travel details, select their route, and choose the type of pass required.

- The proposed system is designed to operate under the control of the state government. All the data, access, and management of the system will be handled by the main State Road Transport Organization of Maharashtra, such as MSRTC. This ensures that the system remains secure, reliable, and officially managed.

  One of the important aspects of this system is that it is scalable and adaptable. Although it is specifically developed for Maharashtra, the same system can be implemented in other states of India by modifying the state name and integrating it with the respective State Transport Department. However, such implementation would require proper approval and authorization from the concerned government authorities.

- The system also improves transparency by allowing students to track the status of their applications. It reduces the need for repeated visits to transport offices and minimizes paperwork. By digitizing the entire process, the proposed system enhances efficiency, accuracy, and user convenience.

  Overall, the Maharashtra Bus Pass System aims to modernize the traditional bus pass process by introducing a centralized, government-controlled digital platform that benefits both students and transport authorities while also providing the flexibility to expand across different states in the future.

2.2 OBJECTIVES OF THE SYSTEM

- To develop an online platform for students to apply for bus passes easily.
  This reduces the need for physical visits and manual form submission.

- To minimize paperwork and make the process fully digital.
  All records are stored securely in a database for easy access.

- To provide a centralized system controlled by the State Transport Authority.
  This ensures proper management and security of all applications.

- To enable administrators to approve or reject applications efficiently.
  This helps in reducing delays and improves overall processing speed.

- To allow students to check the status of their bus pass applications.
  This increases transparency and avoids unnecessary follow-ups.

- To design a system specifically for Maharashtra state students.
  The system can also be adapted for other states with government approval.

- To improve accuracy and reduce human errors in data handling.
  Digital entry ensures proper validation and structured data storage.

- To create a scalable system that can be expanded in the future.
  Features like online payment and QR-based passes can be added later.

2.3 REQUIREMENT GATHERING

Requirement gathering is the foundational phase of the Maharashtra Bus Pass System development lifecycle. This phase involves identifying, analyzing, documenting, and validating the needs and expectations of all stakeholders who will interact with or be affected by the system..

The following requirements were identified:

- **Conductor Feedback** – Input from bus conductors regarding pass validation challenges and desired features for verification.

- **Current System Analysis** – Study of the existing manual bus pass system to identify bottlenecks, delays, and inefficiencies.

- **Document Analysis** – Review of existing application forms, bonafide certificates, and pass formats currently in use.

- **Fare Structure Study** – Analysis of current fare slabs, distance calculations, and concession policies for student passes.

- **Infrastructure Assessment** – Evaluation of available hardware, internet connectivity, and technical infrastructure at institutions and bus depots.

- **Security Requirements** – Gathering requirements for secure authentication, fraud prevention, and counterfeit pass detection.

- **Scalability Needs** – Assessing potential growth in user base and requirement for expansion to other states.

- **Integration Points** – Identifying systems that need integration, including payment gateways, SMS gateways, and institutional databases.

- **Reporting Requirements** – Understanding management reports needed by transport authorities for planning and revenue tracking.

- **Renewal Process** – Understanding current renewal challenges and desired automated renewal features

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2.4 SOFTWARE REQUIREMENT SPECIFICATION (SRS)

The Software Requirement Specifications for the Maharashtra Bus Pass System define the functional and non-functional requirements that the system must fulfill to meet the needs of all stakeholders, including students, educational institutions, transport authorities, and bus conductors.

1. Functional Requirements

- The system should allow students to register and fill the bus pass application form.
- The system should store all user data in a structured database.
- The system should allow administrators to view all applications.
- The system should provide options to approve or reject applications.
- The system should update the application status (Pending / Approved / Rejected).
- The system should allow students to check their application status.
- The system should maintain proper records of all applications.

2. Non-Functional Requirements

- The system should be user-friendly and easy to use.
- The system should provide fast response time for all operations.
- The system should ensure data security and restricted admin access.
- The system should be scalable to support future enhancements.
- The system should be compatible with web browsers and mobile devices.

Chapter 3: System Diagrams

3.1 SYSTEM ARCHITECTURE

The Maharashtra Bus Pass System follows a client-server architecture model. This architecture defines how different components such as user interface, backend processing, and database interact with each other to perform system operations efficiently.
The system mainly consists of three layers:

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1. Presentation Layer (Frontend)
This layer is responsible for user interaction. Students and administrators access the system through a web browser. The interface is designed using HTML and CSS, which provides forms and pages such as application form, status check, and admin dashboard.

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2. Application Layer (Backend)
This layer handles the logic and processing of the system. It is developed using Python Flask. When a user submits data, the backend processes the request, validates input, and performs required operations such as storing data or updating application status.

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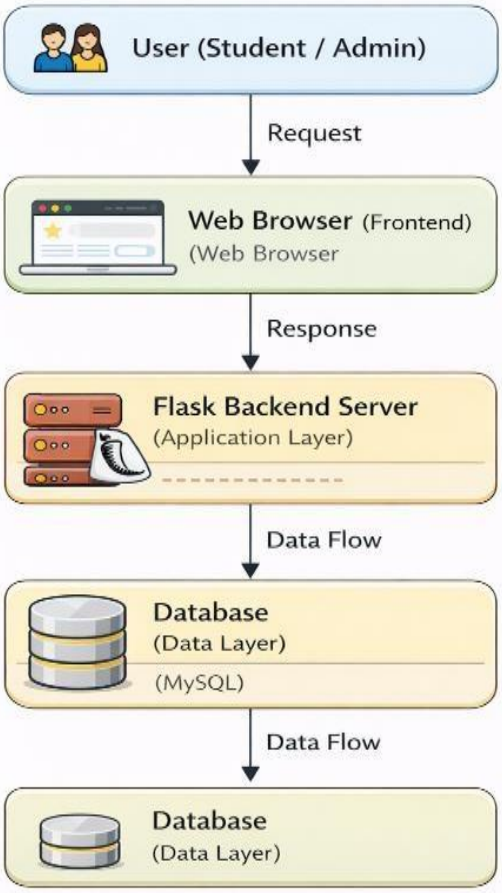
3. Data Layer (Database)
This layer is responsible for storing all system data. A database such as MySQL or SQLite is used to store student details, applications, admin data, and system settings. The backend communicates with the database to store and retrieve data.

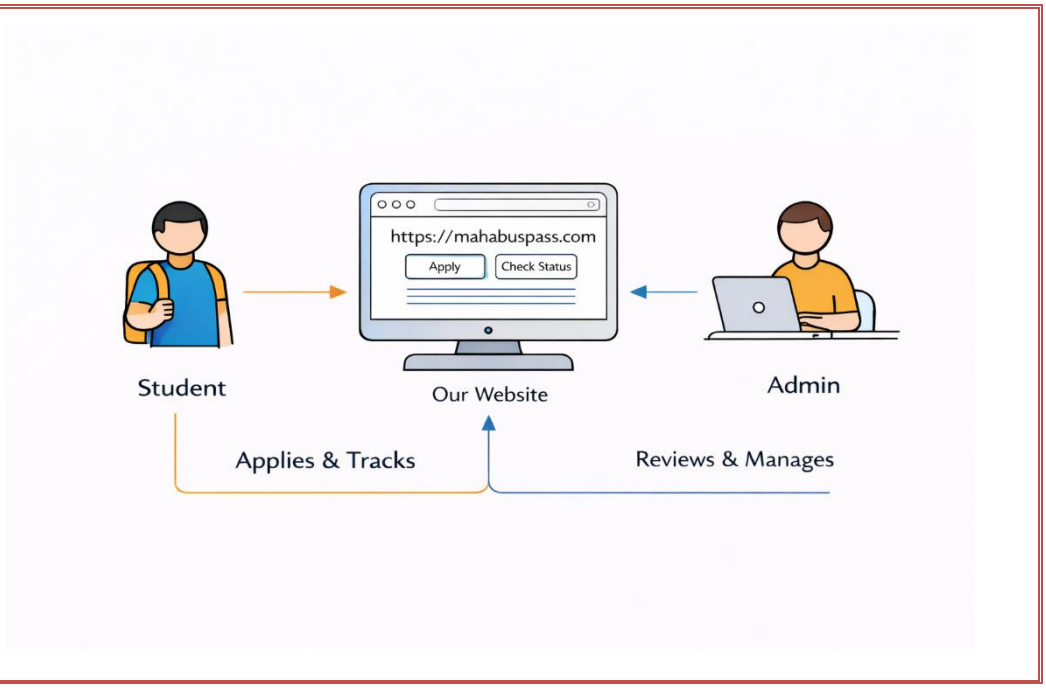
Working of the System Architecture:
• The user (student/admin) sends a request through the web interface.
• The request is processed by the Flask backend server.
• The backend interacts with the database to store or retrieve data.
• The processed result is sent back to the user interface.

Overall, the system architecture ensures smooth communication between different components, efficient data handling, and proper system performance.

System Architecture Diagram



System Architecture Diagram



3.2 DATA FLOW DIAGRAM (DFD)

Context Level DFD (Level 0):
At the context level, the Maharashtra Bus Pass System is represented as a single process interacting with four external entities: Students, Educational Institutions, Transport Authority (MSRTC), and Bus Conductors.

Level 1 DFD:
The Level 1 DFD breaks down the main system into four major processes:

- Process 1: Student Management - handles student registration, login, profile management, and password management.

- Process 2: Application Processing - manages the bus pass application lifecycle including form submission, document upload, application editing, and submission.

- Process 3: Verification and Approval - handles institutional verification, transport authority approval, fare calculation, and pass issuance.

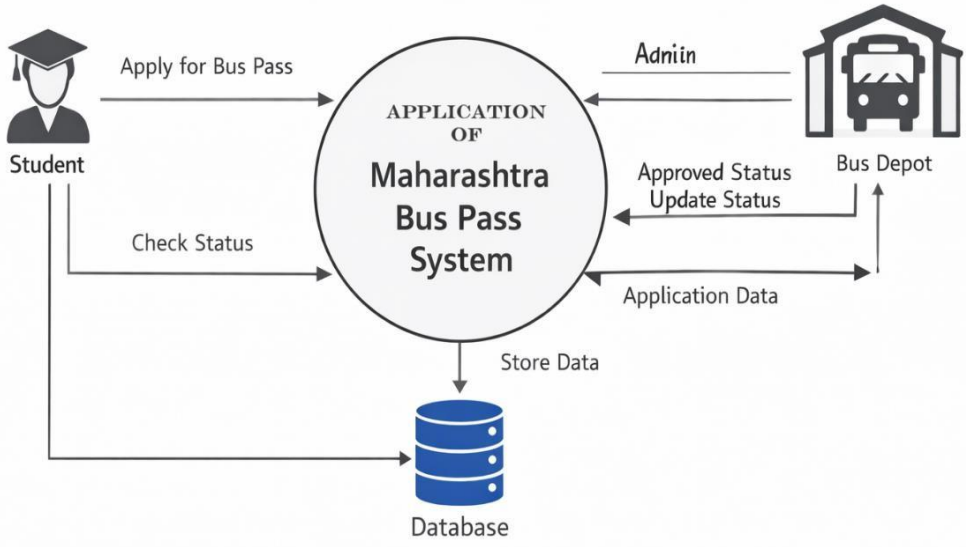
- Process 4: Pass Validation and Management - manages QR code generation, pass validation by conductors, pass renewal, and pass cancellation.

Data stores at Level 1 include: Student Database, Application Database, Institution Database, Pass Database, and Document Storage. Data flows between processes and data stores show how information moves through the system. For example, Process 2 writes application data to the Application Database, while Process 3 reads that data for verification and updates it with approval status.

Level 2 DFD (Application Processing Detailed):

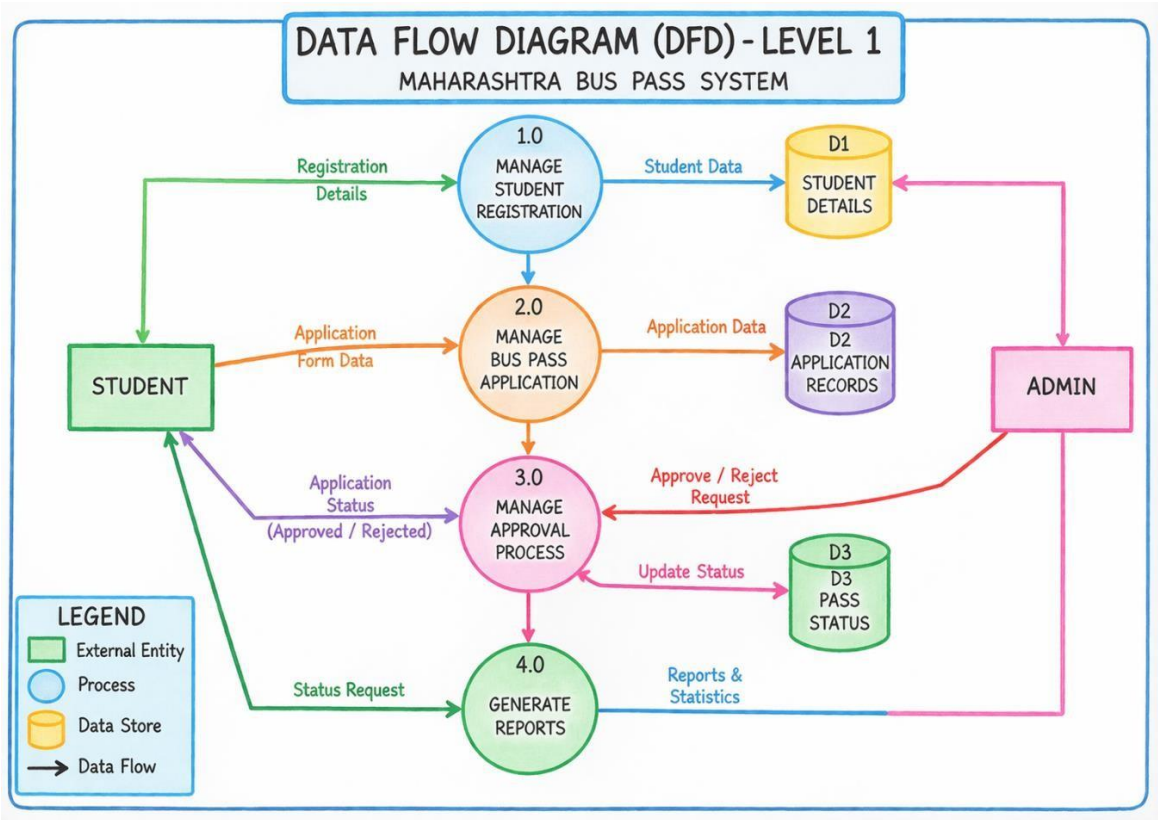
A more detailed Level 2 DFD for the Application Processing process shows sub-processes including:
2.1 - Application Form Filling where students enter personal and route details;
2.2 - Document Upload where students upload required documents with validation checks;
2.3 - Application Preview where students review all entered information;
2.4 - Application Submission where the application is formally submitted and assigned a unique application ID;
2.5 - Application Edit where students can modify pending applications before submission. Data flows between these sub-processes and the Application Database and Document Storage data stores, showing the detailed operations involved in creating a bus pass application.

3.3 DFD LEVEL 0

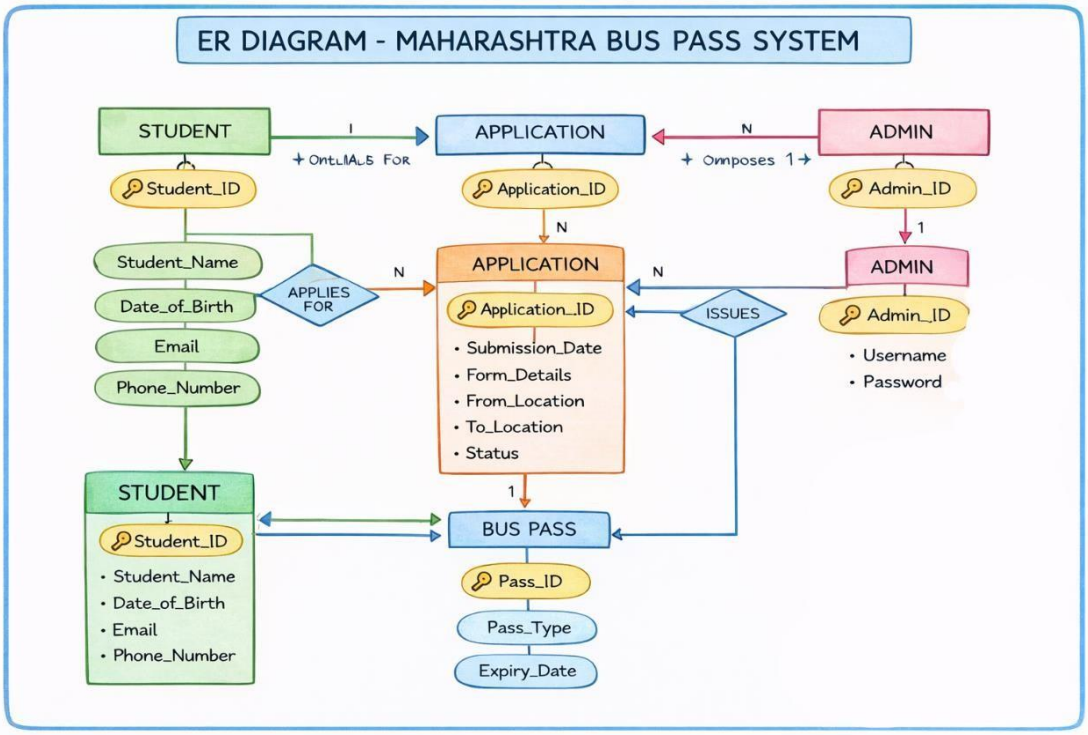


DFD Level 0: Context Diagram

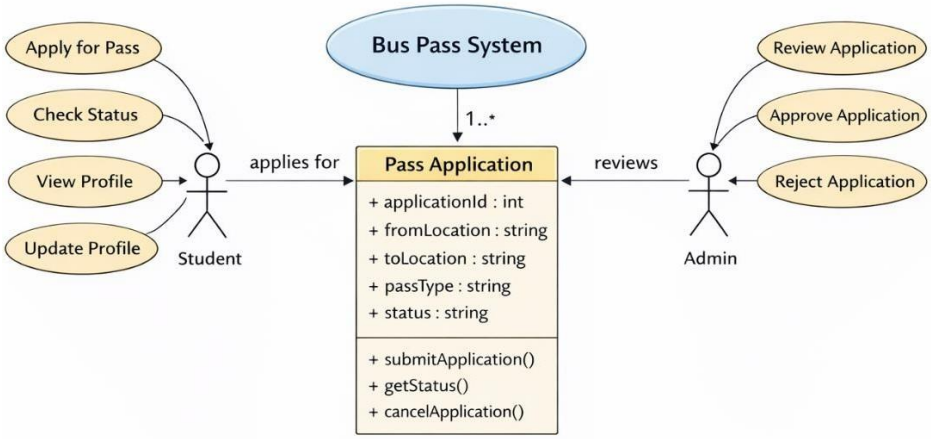
3.4 DFD LEVEL 1



3.5 ENTITY RELATIONSHIP DIAGRAM (ER DIAGRAM)



3.6 UML DIAGRAM



UML DIAGRAM

Chapter 4: System Design

4.1 System Design

The system design of the Maharashtra Bus Pass System describes the overall structure, working, and interaction between different components of the system. This system is designed as a web-based application that helps students apply for bus passes online and allows the State Transport Authority to manage and approve applications efficiently. The design focuses on simplicity, usability, and proper data management.

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1. System Architecture

- The system follows a client-server architecture model.
- The student interacts with the system using a web browser.
- The server processes the request using Python Flask.
- The database is used to store and manage all user and application data.
- All communication between user and server happens through HTTP requests.

---

2. User Interface Design

- The system provides a clean and easy-to-use interface for users.
- It includes different pages such as Home Page, Application Form, Status Check Page, and Admin Dashboard.
- The interface is designed using HTML and CSS for better user experience.
- Buttons, forms, and navigation are kept simple and understandable.
- The design is responsive and works properly on mobile devices and desktops.

---

3. Input Design

- The system collects input from users through an online form.
- Input fields include student name, college name, route details, and pass type.
- All fields are designed to accept only valid data.
- Basic validation is applied to avoid incorrect or incomplete entries.
- The input process is simple so that even non-technical users can use it easily.

---

4. Output Design

- The system provides clear output to the users.
- It displays the application status such as Pending, Approved, or Rejected.
- The admin dashboard shows a list of all applications with details.
- Output is displayed in a readable format with proper alignment.
- The system ensures that users can easily understand the results shown.

---

5. Database Design

- The system uses a database to store all application data.
- A table is created for storing student details and application status.
- Another table is used for admin login credentials.
- Data is stored in a structured format for easy access and management.
- The database ensures that data can be retrieved quickly whenever required.

---

6. Process Design

- The student fills the application form and submits it.
- The system receives the data and stores it in the database.
- The admin logs into the system and views all applications.
- The admin verifies the details and takes action.
- The admin approves or rejects the application.
- The system updates the status accordingly in the database.
- The student can check the updated status anytime.

---

7. Security Design

- The system provides basic security features.
- Admin access is protected using a username and password.
- Only authorized users can access the admin dashboard.
- Data is stored securely in the database.
- The system prevents unauthorized access to sensitive data.

---

8. System Flow

- Student → Fills Application Form
- System → Validates Input Data
- System → Stores Data in Database
- Admin → Reviews Application
- Admin → Approves or Rejects
- System → Updates Status
- Student → Checks Status

---

9. Scalability and Future Design

- The system is designed in a way that it can be expanded in the future.
- It can be modified for other states by changing system configuration.
- Additional features like online payment and QR code can be added.
- Mobile application support can also be developed in future.
- The system can be integrated with government transport services.

Overall, the system design of the Maharashtra Bus Pass System ensures smooth working, easy usability, and efficient management of applications. It provides a structured and scalable solution for digital bus pass management.

4.2 Database Design

The database design of the Maharashtra Bus Pass System is structured using a normalized approach to ensure efficient data management and system performance. The design focuses on maintaining data integrity, reducing redundancy, and enabling fast data retrieval. A well-organized database plays a crucial role in handling user applications, admin operations, and overall system functionality.

The database ensures the following:

• Data is stored securely with proper structure and organization.
• Data redundancy is minimized by avoiding duplicate records.
• Data retrieval is fast and efficient for both users and administrators.
• Relationships between different entities are maintained properly.
• The system supports smooth interaction between application data and admin controls.

The system mainly consists of the following tables:

I. Pass_Applications Table

This table stores all the details related to bus pass applications submitted by students. It includes information such as student details, route information, pass type, and application status (Pending, Approved, or Rejected).

II. Admins Table

This table contains the login credentials of authorized administrators. It ensures that only valid users can access the admin panel and manage applications securely.

III. Students Table

This table stores basic information about students such as name, college details, and identification data. It helps in maintaining user records and linking them with their respective applications.

IV. Settings Table

This table is used to store system-level configurations and settings. It allows flexibility in managing system parameters without modifying the core code.

Overall, the database design provides a strong foundation for the Maharashtra Bus Pass System by ensuring reliable data storage, efficient processing, and smooth system operations.

4.2.1 Pass_applications Table

This table stores all bus pass applications submitted by students, including personal details, route information, document references, and application status.

Field Name	Data Type	Description
id	int (primary key)	Unique id for student from database
student_id	int (foreign key)	Student Id
student_dob	date	Student Date of Birth
unique_pass_id	varchar	Pass Id allotted to student
travel_from	varchar	Starting point of a student
travel_to	varchar	Destination of Student
pass_type	varchar	Monthly, half-yearly, annual
duration	varchar	Pass validity duration days
id_proof_url	varchar	ID proof document path
address_proof_url	varchar	Address proof document path
college_id_url	varchar	College ID document path
fees	decimal	Pass fare amount
payment_status	enum('pending','paid')	Paid, pending, failed
application_status	enum()	Verified, approved, rejected
applied_date	timestamp	Application submission timestamp
approved_date	timestamp	Application approval timestamp
college_id_file	varchar	College ID file path
student_photo_file	varchar	Student photograph file path
address_proof_file	varchar	Address proof file path
bonafide_file	varchar	Bonafide certificate file path
previous_pass_file	varchar	Previous pass file path
address_proof_type	varchar	Aadhaar, passport, license

4.2.2 Admins Table

This table stores all information related to admins of the system administrators.

Field Name	Data Type	Description
id	INT (Primary Key)	Unique ID of admin
username	VARCHAR	Name of admin
password	VARCHAR	Admin password in hashed value

4.2.3 Students Table

This table stores all registered student information, including personal details, educational institution data, and authentication credentials

Field Name	Data Type	Description
id	INT (Primary Key)	Primary key, auto-increment
register_no	VARCHAR	Unique student registration number
name	VARCHAR	Full name of student
email	VARCHAR	Student email address
dob	DATE	Student date of birth
password	VARCHAR	Hashed login password
college	VARCHAR	Educational institution name
course	VARCHAR	Course or class name
year	VARCHAR	Current academic year
student_photo_file	VARCHAR	Student photograph path
phone	VARCHAR	Mobile contact number
address	TEXT	Residential address details
photo_url	VARCHAR	Photograph URL reference
created_at	TIMESTAMP	Account creation timestamp

4.2.4 Settings Table

This table stores system configuration parameters, fare slabs, and customizable settings for the bus pass system.

Field Name	Data Type	Description
id	INT (Primary Key)	Primary key, auto-increment
setting_key	VARCHAR	Unique setting identifier
setting_value	TEXT	Value for the setting
setting_type	VARCHAR	Category of setting
description	TEXT	Brief setting explanation
updated_at	TIMESTAMP	Last modification timestamp

4.3 Input Design

The input design of the Maharashtra Bus Pass System focuses on collecting accurate and relevant data from users in a simple and structured manner. The system is designed in such a way that even non-technical users, such as students, can easily enter their details without any confusion. A well-designed input system helps in reducing errors and ensures smooth processing of applications.

The input design ensures the following:

• The input forms are simple, clear, and user-friendly.
• Only necessary data is collected to avoid complexity.
• Proper validation is applied to ensure correct data entry.
• Input fields are organized in a logical sequence.
• The system minimizes user errors by guiding them during data entry.

The main input in the system is the bus pass application form, which includes the following details:

I. Student Information

This section collects basic details such as the student's full name and college name. These details help in identifying the applicant and linking them with the correct institution.

II. Route Details

The user is required to enter the starting point (From) and destination (To). This information is important for determining the bus route and pass eligibility.

III. Pass Type Selection

The system allows users to select the type of pass, such as Monthly or Quarterly. This helps in defining the validity period of the bus pass.

IV. Form Submission

After entering all the required details, the user submits the form. The system then validates the input data and stores it in the database for further processing.

The input design also includes basic validation techniques such as required fields, proper data types, and controlled input formats. This ensures that incomplete or incorrect data is not submitted into the system.

Overall, the input design of the Maharashtra Bus Pass System plays a vital role in ensuring accurate data collection, reducing user errors, and maintaining the efficiency of the entire system.

4.4 Output Design

The output design of the Maharashtra Bus Pass System focuses on presenting information to users in a clear, simple, and understandable format. The system ensures that both students and administrators receive accurate and meaningful outputs based on their actions. A well-structured output design improves user experience and helps in better decision-making.

The output design ensures the following:

• Information is displayed in a clear and readable format.
• Outputs are organized properly for easy understanding.
• The system provides real-time updates to users.
• Results are shown with proper labels and structure.
• The design avoids confusion by keeping outputs simple and direct.

The main outputs of the system include:

I. Application Status Display

The system shows the current status of the bus pass application, such as Pending, Approved, or Rejected. This helps students to track their application easily.

II. Admin Dashboard Output

The admin dashboard displays a list of all applications along with student details. It helps administrators to review and take necessary actions.

III. Approval and Rejection Updates

When the admin approves or rejects an application, the updated status is reflected immediately in the system. This ensures transparency.

IV. System Messages

The system displays messages such as successful submission, errors, or updates. These messages guide users during their interaction with the system.

V. Student Dashboard

The system shows the status of the Student pass application and other his data such as Pending, Approved, Rejected Pass and his other data. This helps students to track their application easily.

Overall, the output design ensures that all system responses are clear, meaningful, and useful for both students and administrators.

4.5 Module Design

The Maharashtra Bus Pass System is divided into different modules to simplify the overall functionality and improve system organization. Each module performs a specific task and works together with other modules to complete the system operations efficiently.

The main modules of the system are:

I. Student Module

This module allows students to apply for a bus pass by filling out the application form. It also enables students to check the status of their application.

II. Admin Module

This module is accessible only to authorized users. It allows the admin to view all applications, verify details, and approve or reject requests.

III. Application Management Module

This module handles the processing of application data. It manages the flow of data from submission to approval.

IV. Database Module

This module is responsible for storing and retrieving all system data. It ensures that data is managed in a structured and efficient manner.

V. Authentication Module

This module ensures that only authorized users can access the admin panel. It provides login security and access control.

Overall, the modular design helps in improving system efficiency, maintainability, and scalability.

4.6 Security Design

The security design of the Maharashtra Bus Pass System ensures that the system is protected from unauthorized access and data misuse. It focuses on maintaining the confidentiality, integrity, and availability of data.

The security design includes the following features:

• Admin access is protected using username and password authentication.
• Only authorized users can access sensitive system data.
• Input validation is applied to prevent incorrect or harmful data entry.
• The system prevents unauthorized modification of data.
• Basic security measures are implemented to protect user information.

Security mechanisms used in the system:

I. Authentication

The system verifies the identity of the admin using login credentials before granting access.

II. Authorization

Only authorized users are allowed to perform admin operations such as approving or rejecting applications.

III. Data Protection

All user data is stored securely in the database to prevent data loss or misuse.

IV. Input Validation

The system checks user inputs to avoid invalid or harmful data submission.

Overall, the security design ensures that the Maharashtra Bus Pass System remains safe, reliable, and protected from potential threats.

CHAPTER 5: USER GUIDELINES

5. USER GUIDELINES

The Maharashtra Bus Pass System is designed to provide a simple and user-friendly platform for students and administrators. This section provides clear instructions on how to use the system effectively. It ensures that users can easily understand the process of applying for a bus pass and managing applications without any confusion.

5.1 Guidelines for Students

Students are the primary users of this system. They can apply for bus passes and check the status of their applications. The following steps should be followed:

I. Accessing the System

The student should open the website using a web browser such as Chrome or Edge. A stable internet connection is required for proper functioning.

II. Filling the Application Form

The student must click on the "Apply for Bus Pass" option. All required details such as name, college name, route (From – To), and pass type should be entered correctly.

III. Providing Accurate Information

Students must ensure that all the information provided is correct. Incorrect or incomplete data may lead to rejection of the application.

IV. Submitting the Application

After filling all details, the student should click on the submit button. The system will store the application in the database.

V. Checking Application Status

Students can check their application status by visiting the status page. The status may be shown as Pending, Approved, or Rejected.

VI. Following Rules and Conditions

Students must follow all rules set by the transport authority. Misuse of the system or providing false information may lead to cancellation of the pass.

5.2 Guidelines for Administrators

Administrators are responsible for managing and approving applications. They have full control over the system.

I. Admin Login

The admin must log in using valid username and password credentials. Unauthorized users are not allowed to access the admin panel.

II. Viewing Applications

The admin dashboard displays all submitted applications along with student details. The admin should review each application carefully.

III. Verifying Details

The admin must verify the information provided by the student before taking any action. This ensures the authenticity of applications.

IV. Approving or Rejecting Applications

The admin can approve or reject applications based on the verification process. Once updated, the status is reflected in the system.

V. Maintaining Records

The admin should maintain proper records of all applications for future reference. This helps in tracking and reporting.

5.3 General Guidelines

These guidelines apply to both students and administrators for smooth system operation:

I. Use Valid Information
All users must provide correct and complete information while using the system.

II. Avoid Multiple Submissions
Students should avoid submitting multiple applications for the same pass.

III. Maintain System Security
Admin login credentials should be kept confidential to prevent unauthorized access.

IV. Follow System Instructions
Users should carefully read and follow all instructions displayed on the website.

V. Technical Requirements
The system works best on modern web browsers with an active internet connection.

5.4 Important Notes

• The system is designed specifically for Maharashtra State students.
• The system operates under the control of the State Road Transport Authority.
• Any misuse of the system may result in rejection or cancellation of the application.
• Future updates may include additional features and improvements.

5.5 System Usage Instructions

This section provides step-by-step instructions for using the Maharashtra Bus Pass System effectively. It helps users understand how to navigate through the system and perform required actions smoothly.

I. Starting the System
The user should open the system using a web browser and access the home page of the application.

II. Navigating the Website
Users can move between different pages such as Home, Apply, Status, and Admin using the navigation options provided on the website.

III. Performing Actions
Students can apply for a bus pass, while administrators can log in and manage applications using the dashboard.

IV. Viewing Results
After performing actions, users can view results such as application status or updated records on the screen.

V. Logging Out (Admin)
Administrators should log out after completing their work to maintain system security.

Overall, these instructions help users to operate the system efficiently without confusion and ensure proper usage of all features.

Overall, the user guidelines ensure that both students and administrators can use the Maharashtra Bus Pass System efficiently and correctly. It helps in maintaining system discipline, improving user experience, and ensuring smooth operation.

CHAPTER 6: SOURCE CODE

6.1 : Main Module: (app.py)

```
from flask import Flask, render_template, request, redirect, url_for, session, jsonify, flash
import pymysql
import pymysql.cursors
import re
import os
import random
import string
from datetime import datetime
from werkzeug.utils import secure_filename

app = Flask(__name__, static_folder='static')

# ---------- JINJA2 GLOBALS ----------
app.jinja_env.globals.update(now=datetime.now)

# ---------- SECRET KEY ----------
app.secret_key = 'bus_pass_secret_key_2024'

# ---------- FILE UPLOAD CONFIG ----------
app.config['UPLOAD_FOLDER'] = 'static/uploads'
app.config['MAX_CONTENT_LENGTH'] = 5 * 1024 * 1024
app.config['ALLOWED_EXTENSIONS'] = {'jpg', 'jpeg', 'png', 'pdf'}

os.makedirs(app.config['UPLOAD_FOLDER'], exist_ok=True)

def allowed_file(filename):
    return '.' in filename and \
           filename.rsplit('.', 1)[1].lower() in app.config['ALLOWED_EXTENSIONS']

# ---------- DATABASE CONFIG ----------
db_config = {
    'host': 'localhost',
    'user': 'root',
    'password': ' ',
    'database': 'bus_pass_system',
    'cursorclass': pymysql.cursors.DictCursor
}

def get_db_connection():
    return pymysql.connect(**db_config)

@app.route('/submit_application', methods=['POST'])
def submit_application():
    if 'loggedin' not in session:
        return redirect(url_for('login'))

    try:
        travel_from = request.form['travel_from']
        travel_to = request.form['travel_to']
        pass_type = request.form['pass_type']
        duration = request.form['duration']
        address_proof_type = request.form.get('address_proof_type', '')

        if duration == '1 Month':
            fees = 500
        elif duration == '3 Months':
            fees = 1500
        else:
```

```
            fees = 3000

        unique_pass_id = generate_pass_id()

        college_id_file = None
        student_photo_file = None
        address_proof_file = None
        previous_pass_file = None
        bonafide_file = None

        if 'bonafide' in request.files:
            file = request.files['bonafide']
            if file and file.filename:
                if allowed_file(file.filename):
                    timestamp = datetime.now().strftime('%Y%m%d%H%M%S')
                    filename = secure_filename(f"bonafide_{session['student_id']}_{timestamp}_{file.filename}")
                    file_path = os.path.join(app.config['UPLOAD_FOLDER'], filename)
                    file.save(file_path)
                    bonafide_file = filename

        if 'college_id' in request.files:
            file = request.files['college_id']
            if file and file.filename:
                if allowed_file(file.filename):
                    timestamp = datetime.now().strftime('%Y%m%d%H%M%S')
                    filename =
secure_filename(f"college_id_{session['student_id']}_{timestamp}_{file.filename}")
                    file.save(os.path.join(app.config['UPLOAD_FOLDER'], filename))
                    college_id_file = filename

        connection = get_db_connection()
        try:
            with connection.cursor() as cursor:
                sql = '''INSERT INTO pass_applications
                        (student_id, unique_pass_id, travel_from, travel_to, pass_type,
                         duration, fees, college_id_file, student_photo_file,
                         address_proof_type, address_proof_file, previous_pass_file, bonafide_file,
                         application_status, payment_status)
                        VALUES (%s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s, %s)'''
                cursor.execute(sql, (
                    session['student_id'], unique_pass_id, travel_from, travel_to,
                    pass_type, duration, fees + 50,
                    college_id_file, student_photo_file,
                    address_proof_type, address_proof_file, previous_pass_file, bonafide_file,
                    'pending', 'pending'
                ))
                connection.commit()
                application_id = cursor.lastrowid
        except Exception as e:
            print(f"Database error: {e}")
            connection.rollback()
            raise e
        finally:
            connection.close()

        return redirect(url_for('payment', application_id=application_id))

    except Exception as e:
        print(f"Error in submit_application: {e}")
        import traceback
        traceback.print_exc()
        return "Error processing application. Check console for details.", 500
```

```
# ---------- ADMIN ROUTES ----------
@app.route('/admin/login', methods=['GET', 'POST'])
def admin_login():
    if request.method == 'POST':
        username = request.form['username']
        password = request.form['password']

        if username == 'admin' and password == 'admin123':
            session['admin_loggedin'] = True
            return redirect(url_for('admin_dashboard'))
        else:
            return render_template('admin_login.html', error='Invalid credentials')

    return render_template('admin_login.html')

@app.route('/admin/dashboard')
def admin_dashboard():
    if 'admin_loggedin' in session:
        connection = get_db_connection()
        try:
            with connection.cursor() as cursor:
                cursor.execute('''
                    SELECT pa.*, s.name, s.register_no, s.college
                    FROM pass_applications pa
                    JOIN students s ON pa.student_id = s.id
                    ORDER BY pa.applied_date DESC
                ''')
                applications = cursor.fetchall()

                cursor.execute('SELECT COUNT(*) as total FROM pass_applications')
                total_applications = cursor.fetchone()['total']

                cursor.execute('SELECT COUNT(*) as approved FROM pass_applications WHERE
application_status = "approved"')
                approved_count = cursor.fetchone()['approved']

                cursor.execute('SELECT COUNT(*) as pending FROM pass_applications WHERE
application_status = "pending"')
                pending_count = cursor.fetchone()['pending']

                cursor.execute('SELECT COUNT(*) as rejected FROM pass_applications WHERE
application_status = "rejected"')
                rejected_count = cursor.fetchone()['rejected']

                cursor.execute('SELECT COUNT(*) as paid FROM pass_applications WHERE payment_status
= "paid"')
                paid_count = cursor.fetchone()['paid']

                cursor.execute('SELECT COUNT(*) as pending_payment FROM pass_applications WHERE
payment_status = "pending"')
                pending_payment_count = cursor.fetchone()['pending_payment']

                cursor.execute('SELECT COUNT(*) as today FROM pass_applications WHERE
DATE(applied_date) = CURDATE()')
                today_applications = cursor.fetchone()['today']

                cursor.execute('SELECT COUNT(*) as total FROM students')
                total_students = cursor.fetchone()['total']

        except Exception as e:
```

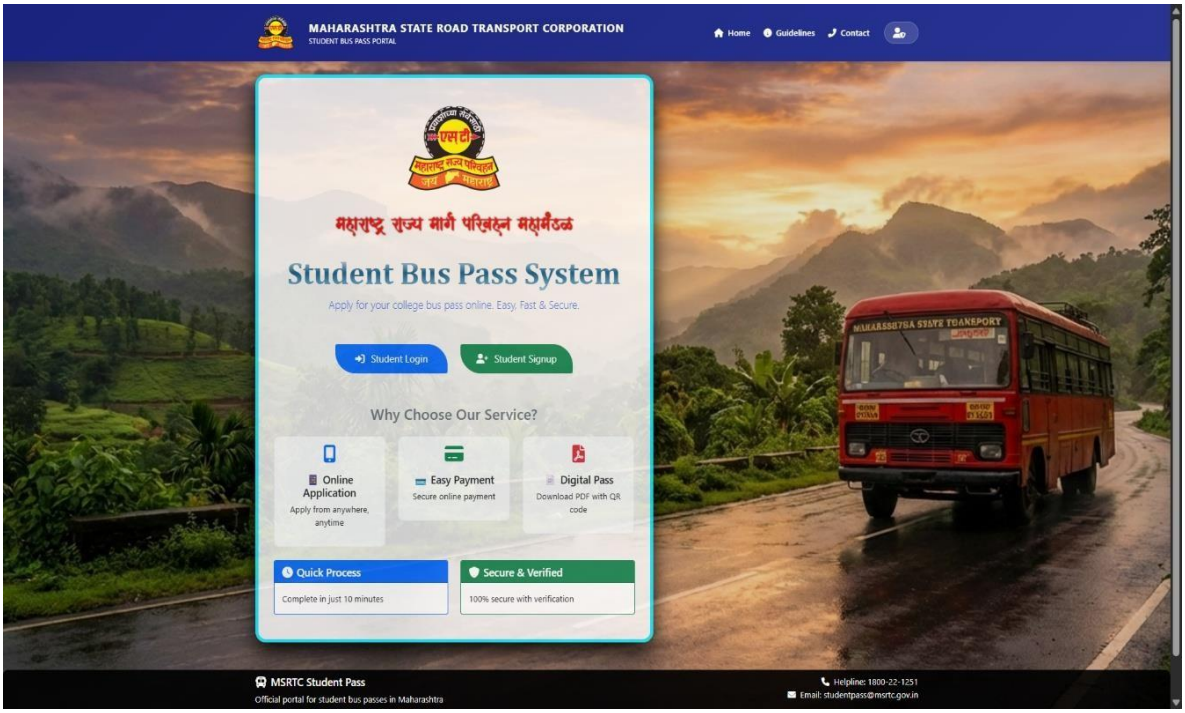
```
            print(f"Admin dashboard error: {e}")
            applications = []
            total_applications = approved_count = pending_count = rejected_count = paid_count =
pending_payment_count = today_applications = total_students = 0
        finally:
            connection.close()

        return render_template('admin_dashboard.html',
                               applications=applications,
                               total_applications=total_applications,
                               approved_count=approved_count,
                               pending_count=pending_count,
                               rejected_count=rejected_count,
                               paid_count=paid_count,
                               pending_payment_count=pending_payment_count,
                               today_applications=today_applications,
                               total_students=total_students)
    return redirect(url_for('admin_login'))

@app.route('/admin/all_students')
def admin_all_students():
    if 'admin_loggedin' in session:
        connection = get_db_connection()
        try:
            with connection.cursor() as cursor:
                cursor.execute('''
                    SELECT
                        s.*,
                        COUNT(pa.id) as total_applications,
                        SUM(CASE WHEN pa.application_status = 'approved' THEN 1 ELSE 0 END) as
approved_applications,
                        SUM(CASE WHEN pa.application_status = 'pending' THEN 1 ELSE 0 END) as
pending_applications,
                        MAX(pa.applied_date) as last_application_date
                    FROM students s
                    LEFT JOIN pass_applications pa ON s.id = pa.student_id
                    GROUP BY s.id
                    ORDER BY s.id DESC
                ''')
                students = cursor.fetchall()

                cursor.execute('SELECT COUNT(*) as total FROM students')
                total_students = cursor.fetchone()['total']

                cursor.execute('SELECT COUNT(DISTINCT student_id) as active FROM pass_applications')
                active_students = cursor.fetchone()['active']

        except Exception as e:
            print(f"Error in admin_all_students: {e}")
            students = []
            total_students = 0
            active_students = 0
        finally:
            connection.close()

        return render_template('admin_all_students.html',
                               students=students,
                               total_students=total_students,
                               active_students=active_students)
    return redirect(url_for('admin_login'))

@app.route('/admin/applications_remains')
```

```
def admin_applications_remains():
    if 'admin_loggedin' in session:
        connection = get_db_connection()
        try:
            with connection.cursor() as cursor:
                cursor.execute('''
                    SELECT
                        pa.*,
                        s.name,
                        s.register_no,
                        s.college,
                        s.course,
                        s.year,
                        s.email,
                        s.phone,
                        s.address
                    FROM pass_applications pa
                    JOIN students s ON pa.student_id = s.id
                    WHERE pa.application_status = 'pending'
                    AND pa.payment_status = 'paid'
                    ORDER BY pa.applied_date ASC
                ''')
                pending_applications = cursor.fetchall()

                cursor.execute('''
                    SELECT
                        COUNT(*) as total_pending,
                        COUNT(DISTINCT student_id) as unique_students,
                        SUM(CASE WHEN DATE(pa.applied_date) = CURDATE() THEN 1 ELSE 0 END) as
today_applications
                    FROM pass_applications pa
                    WHERE pa.application_status = 'pending'
                    AND pa.payment_status = 'paid'
                ''')
                stats = cursor.fetchone()

                for app in pending_applications:
                    app['documents_verified'] = all([
                        app.get('college_id_file'),
                        app.get('student_photo_file'),
                        app.get('address_proof_file')
                    ])

        except Exception as e:
            print(f"Error in admin_applications_remains: {e}")
            pending_applications = []
            stats = {'total_pending': 0, 'unique_students': 0, 'today_applications': 0}
        finally:
            connection.close()

        return render_template('admin_applications_remains.html',
                               applications=pending_applications,
                               stats=stats)
    return redirect(url_for('admin_login'))

# ---------- ADMIN SETTINGS ----------

@app.route('/admin/settings/reset', methods=['POST'])
def admin_settings_reset():
    if 'admin_loggedin' not in session:
        return redirect(url_for('admin_login'))
```

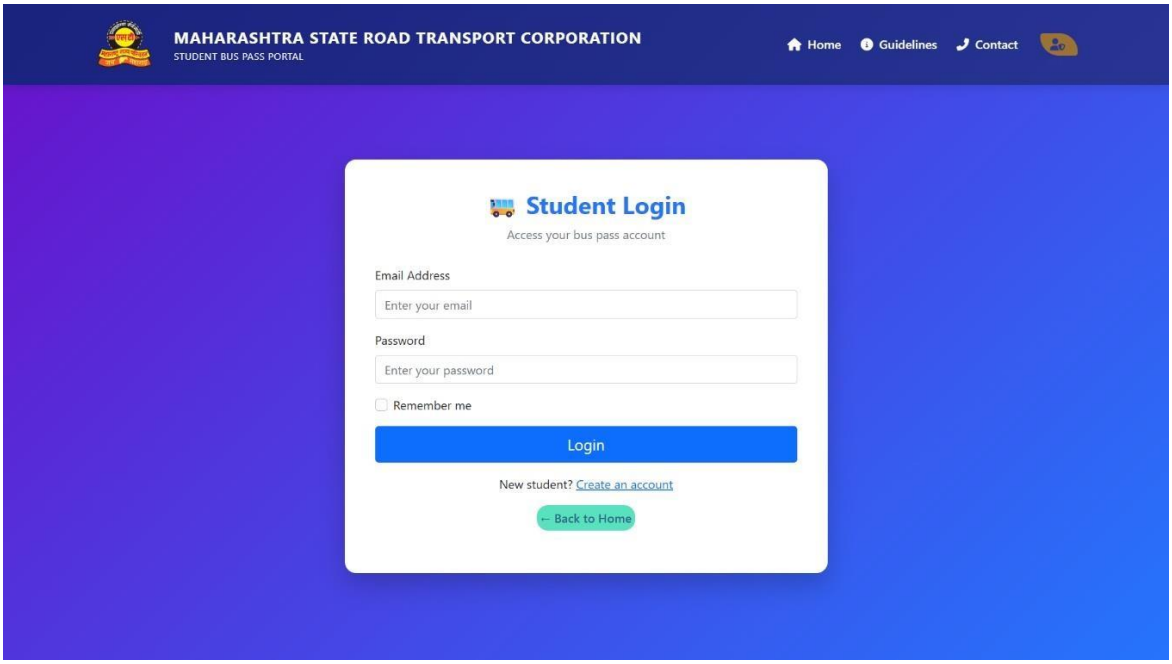
```
    connection = get_db_connection()
    try:
        with connection.cursor() as cursor:
            cursor.execute('DELETE FROM settings')
            reset_default_settings(cursor)
            connection.commit()
        flash('Settings reset to defaults!', 'success')
    except Exception as e:
        flash(f'Reset failed: {str(e)}', 'danger')
    finally:
        connection.close()

    return redirect(url_for('admin_settings'))

def reset_default_settings(cursor):
    defaults = [
        ('system_name', 'Bus Pass Management System', 'text'),
        ('system_email', 'admin@buspass.com', 'email'),
        ('system_phone', '+91 1234567890', 'text'),
        ('system_address', 'College Campus, City', 'textarea'),
        ('processing_fee', '50', 'number'),
        ('pass_regular_1month', '500', 'number'),
        ('pass_regular_3months', '1500', 'number'),
        ('pass_regular_6months', '3000', 'number'),
        ('pass_premium_1month', '800', 'number'),
        ('pass_premium_3months', '2300', 'number'),
        ('pass_premium_6months', '4500', 'number'),
        ('pass_express_1month', '600', 'number'),
        ('pass_express_3months', '1750', 'number'),
        ('pass_express_6months', '3400', 'number'),
        ('max_file_size', '2', 'number'),
        ('allowed_extensions', 'jpg,jpeg,png,pdf', 'text'),
        ('require_previous_pass', '0', 'boolean'),
        ('enable_email_notifications', '1', 'boolean'),
        ('enable_sms_notifications', '0', 'boolean'),
        ('smtp_host', '', 'text'),
        ('smtp_port', '587', 'number'),
        ('smtp_user', '', 'text'),
        ('smtp_password', '', 'password'),
        ('auto_approve', '0', 'boolean'),
        ('require_verification', '1', 'boolean'),
        ('student_can_edit', '0', 'boolean'),
        ('max_applications_per_day', '3', 'number'),
        ('items_per_page', '10', 'number'),
        ('theme_color', 'primary', 'select'),
        ('date_format', 'd/m/Y', 'text'),
    ]
    for key, value, typ in defaults:
        cursor.execute(
            'INSERT INTO settings (setting_key, setting_value, setting_type) VALUES (%s, %s, %s)',
            (key, value, typ)
        )

@app.route('/admin/settings/cache/clear', methods=['POST'])
def admin_settings_clear_cache():
    if 'admin_loggedin' in session:
        flash('Cache cleared successfully!', 'success')
    return redirect(url_for('admin_settings'))

# ------- MAIN -------
if __name__== '__main__':
    app.run(debug=True)
```

Chapter 7: Outputs

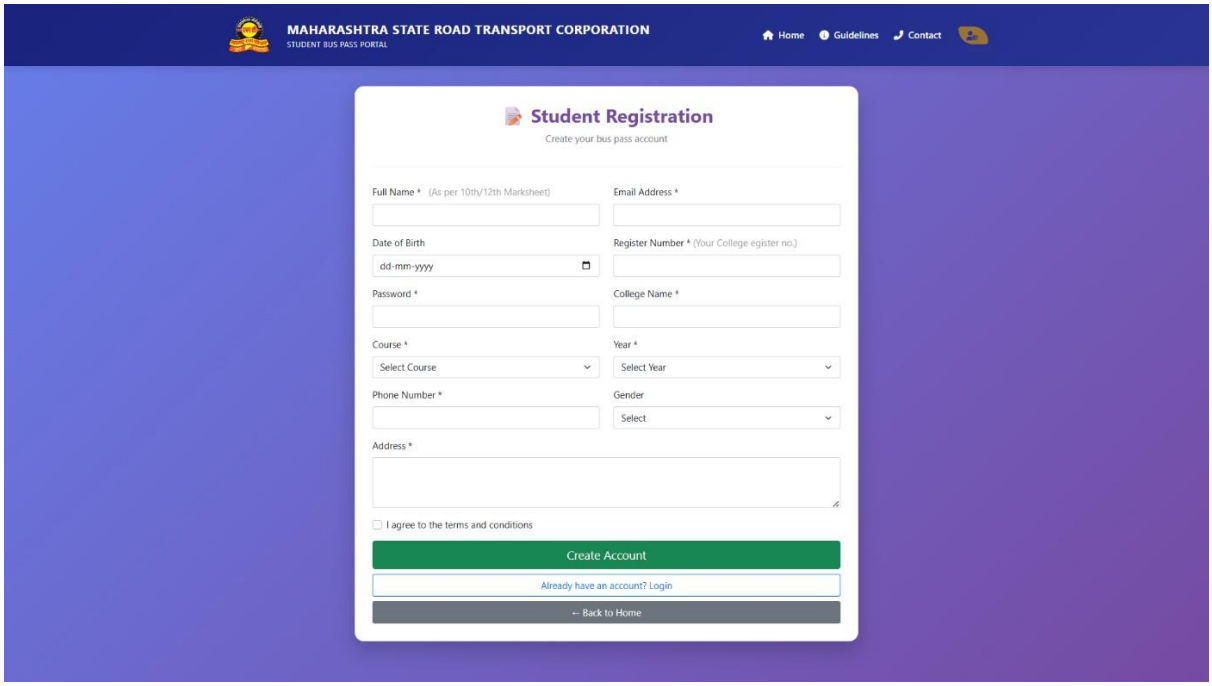
7.1 Index Page



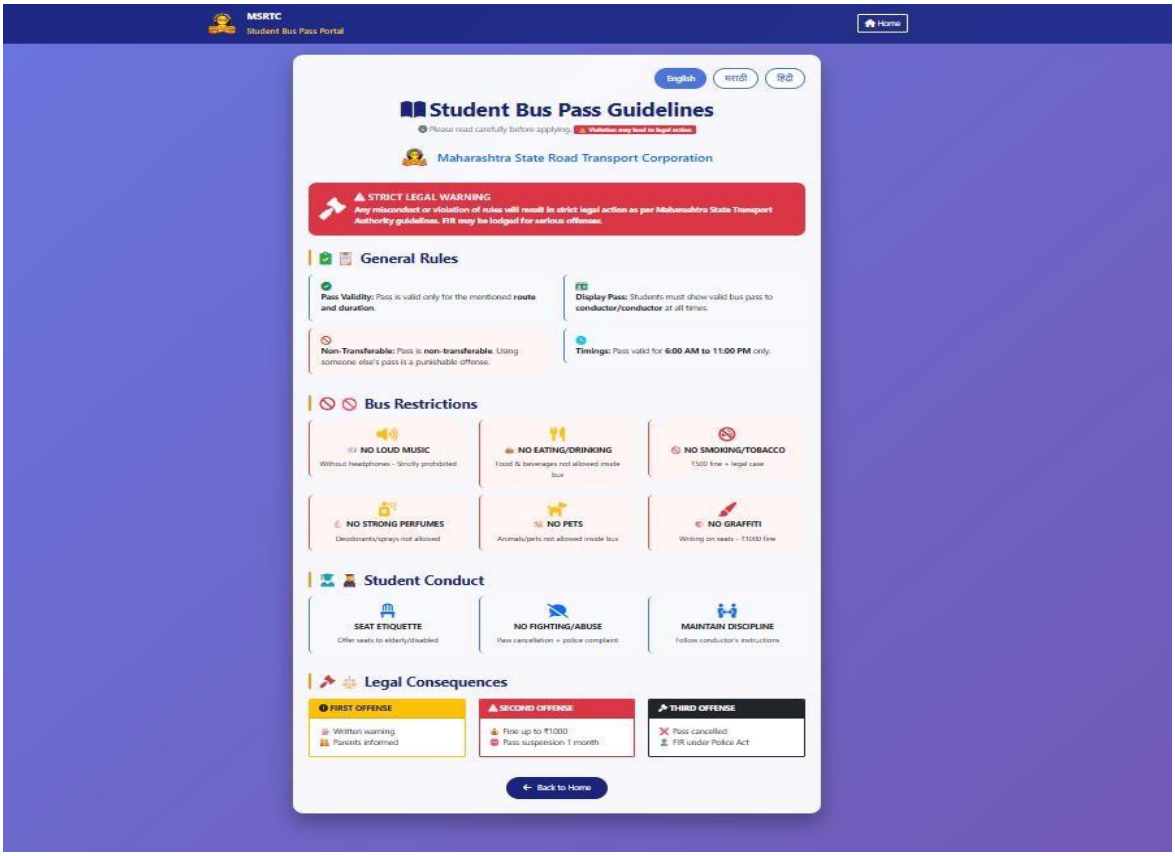
7.2 Login page



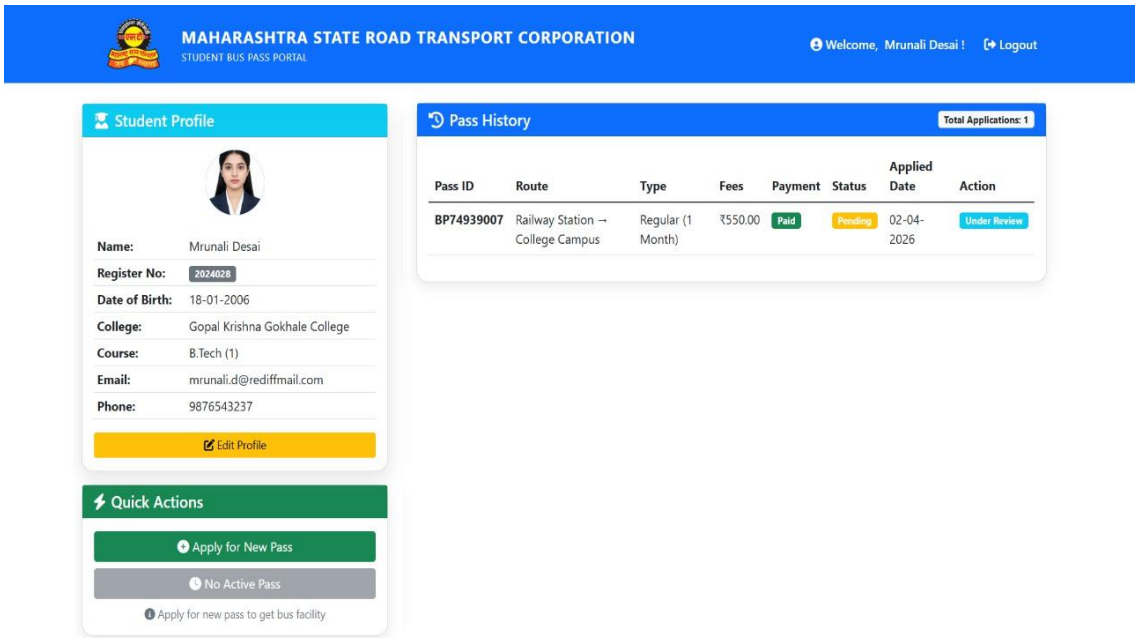
7.3 Sign-up page



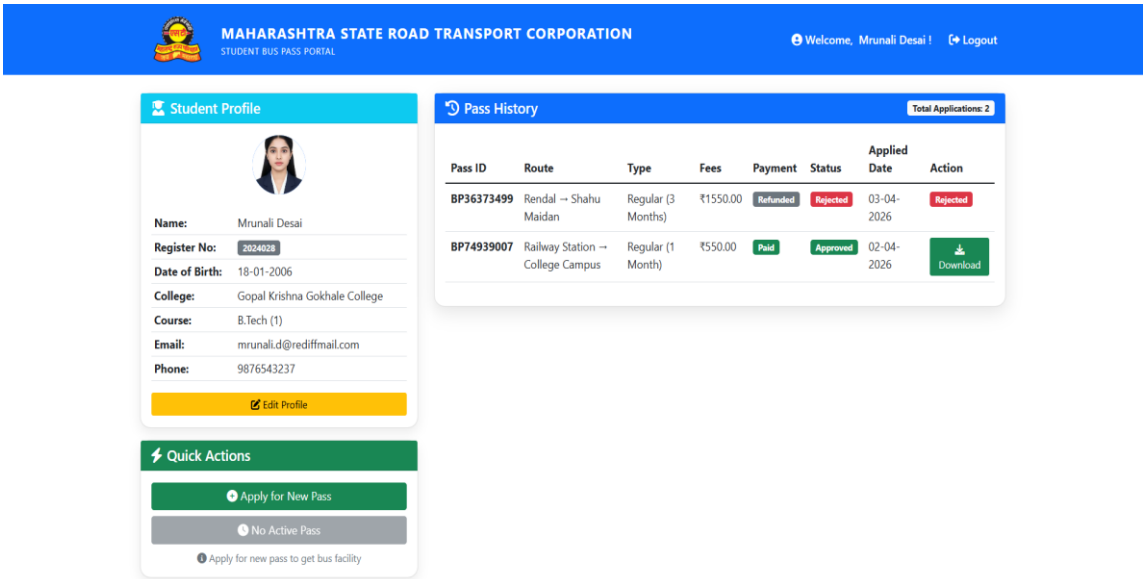
7.4 Guidelines of Bus



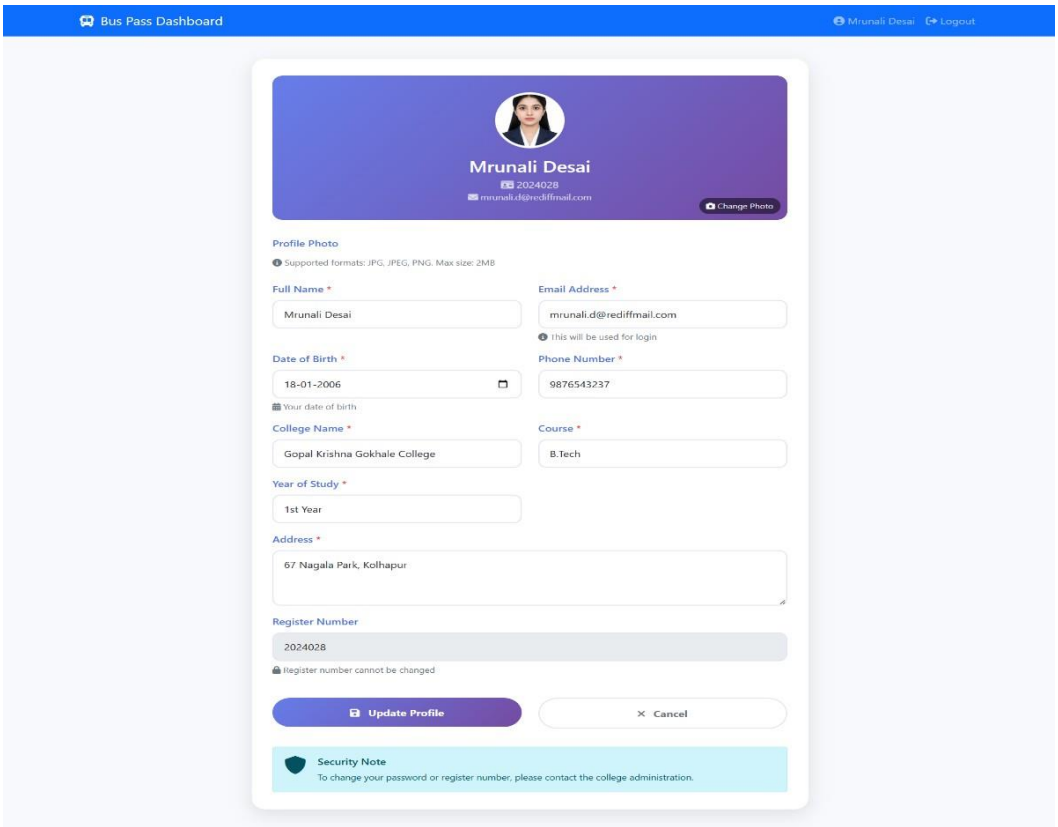
7.5 Student Dashboard after applying for pass



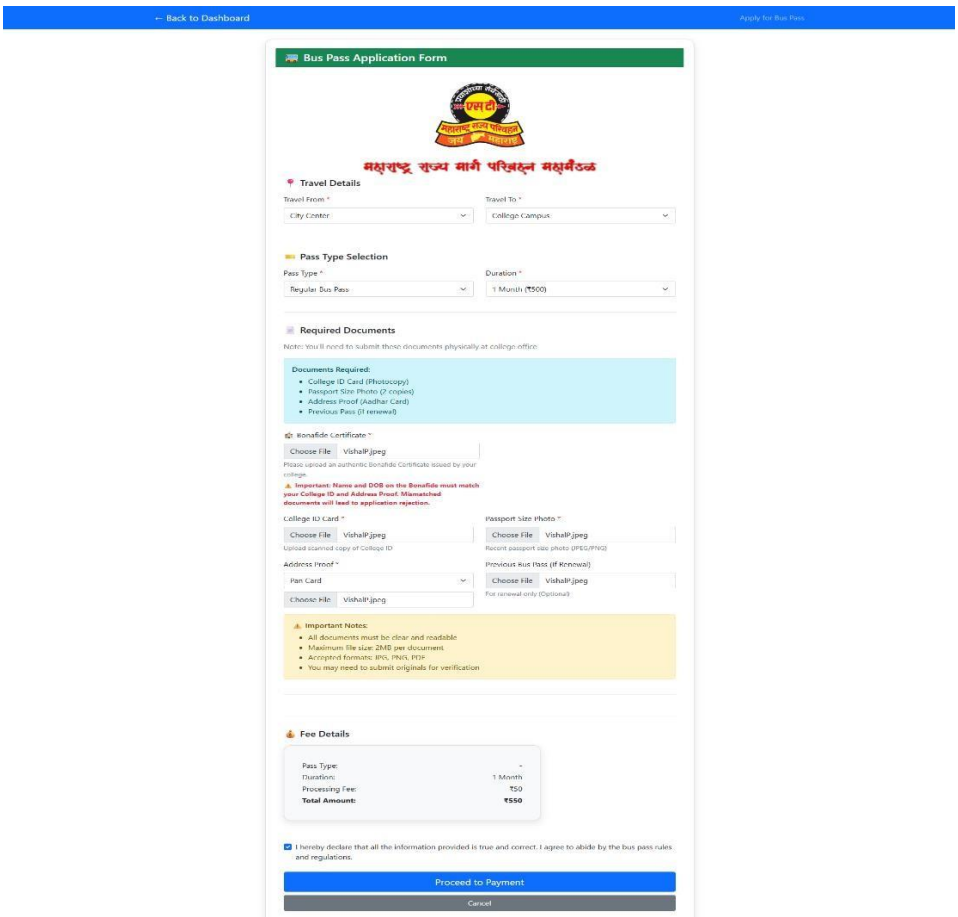
- **Student Dashboard after Approval or Rejection**



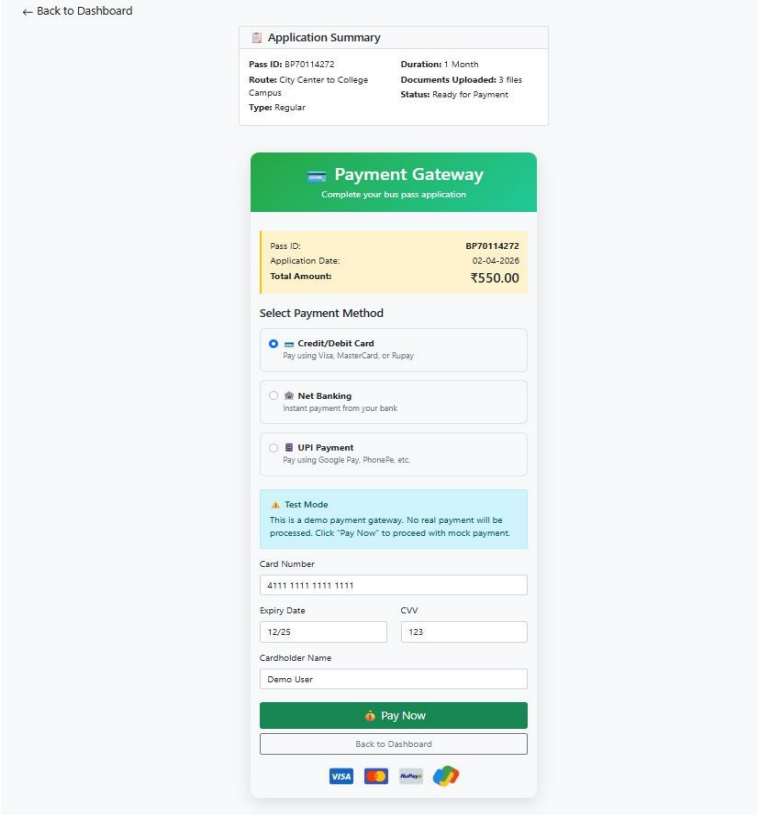
- **Student Can Update their Profile**



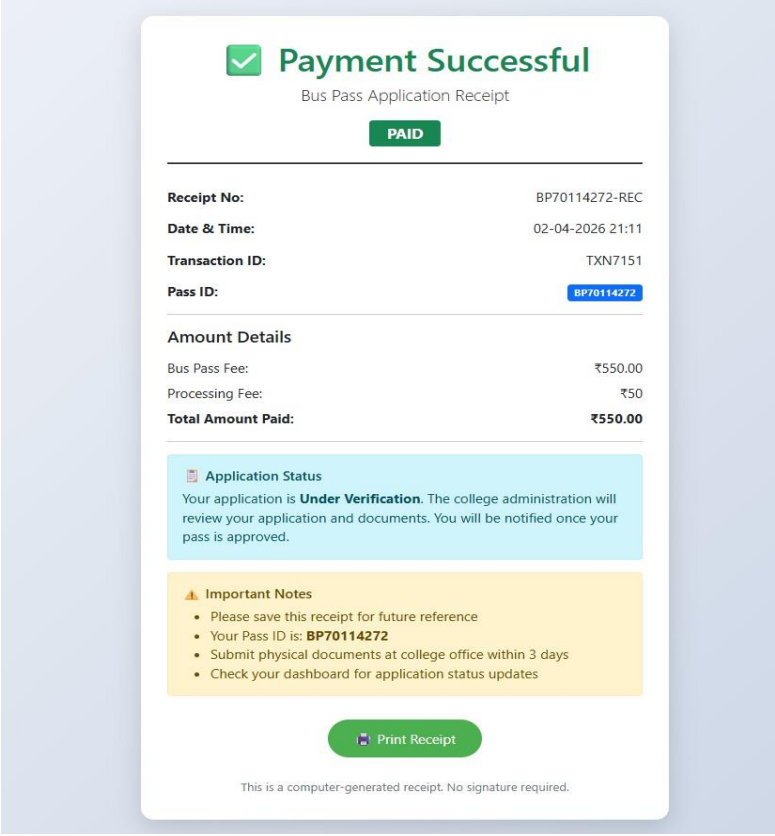
7.6 Application For Pass



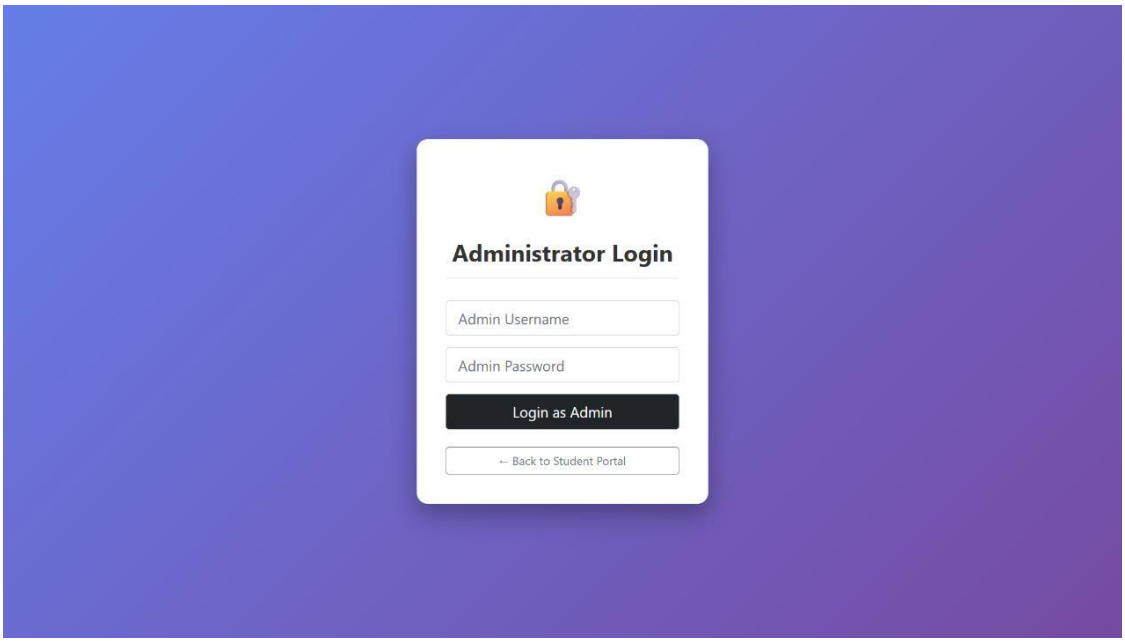
7.7 Payment page



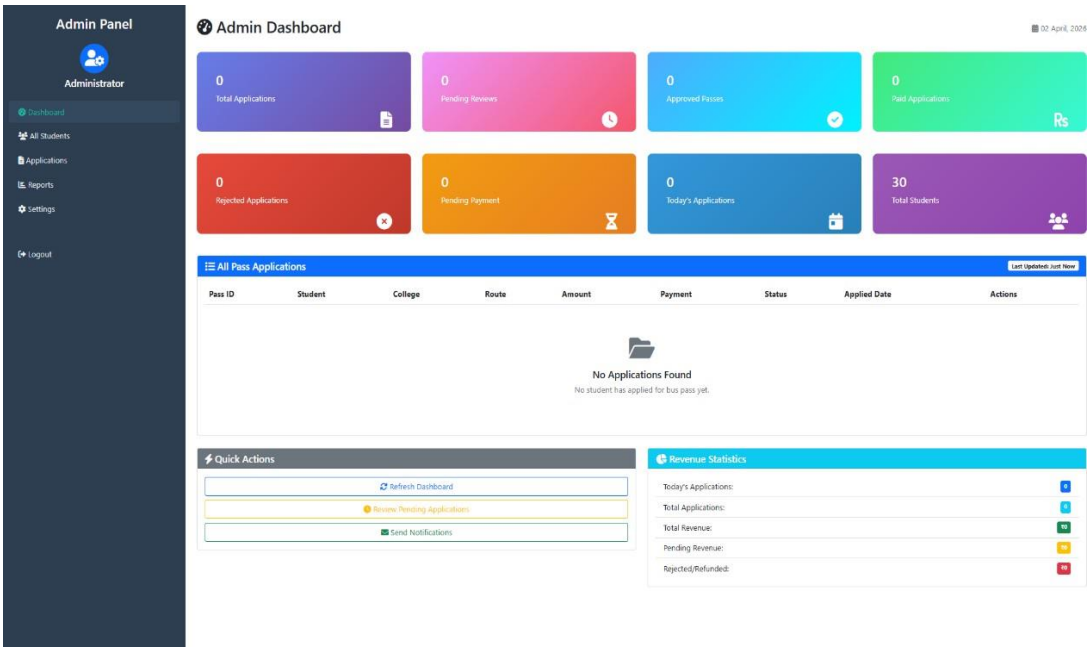
- **Printable Payment Receipt**



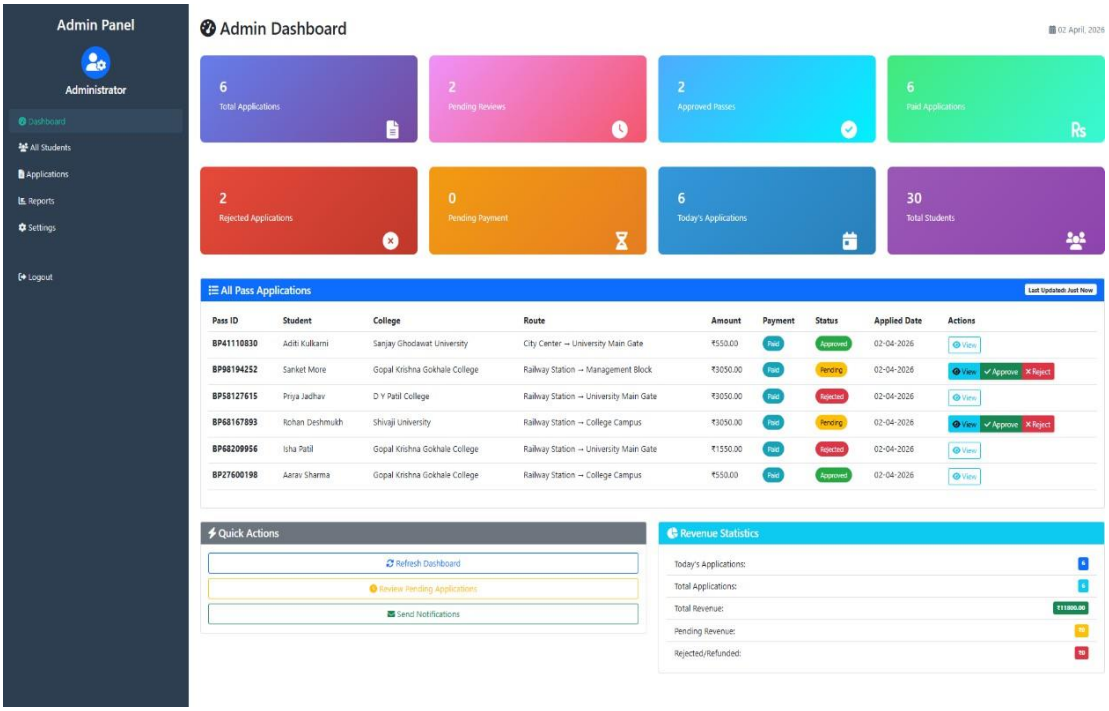
7.8 Admin Login



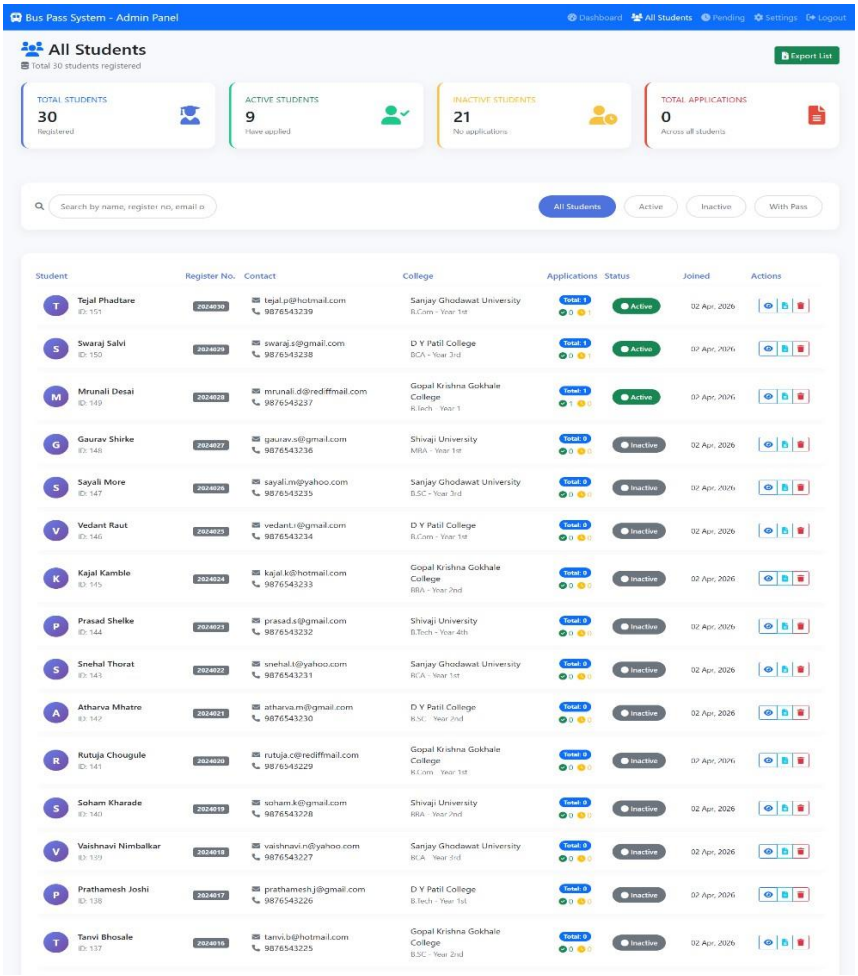
- **Empty Admin Dashboard (when no records found)**



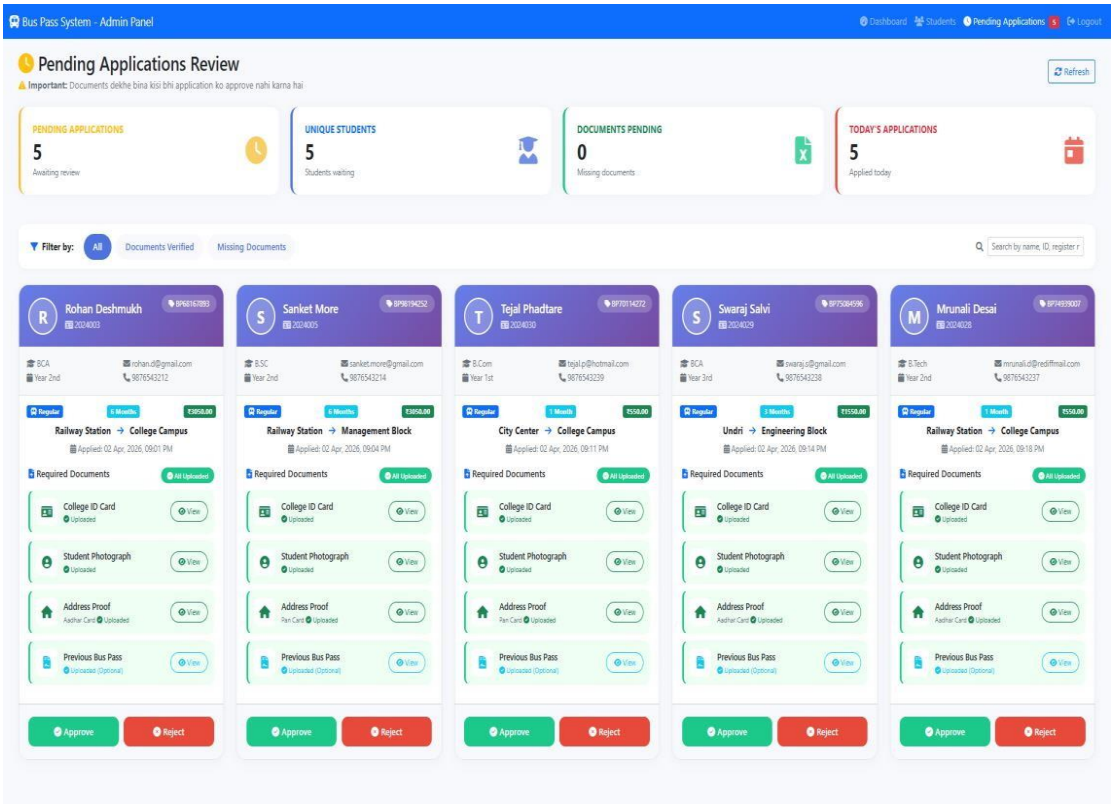
- **Admin Dashboard**



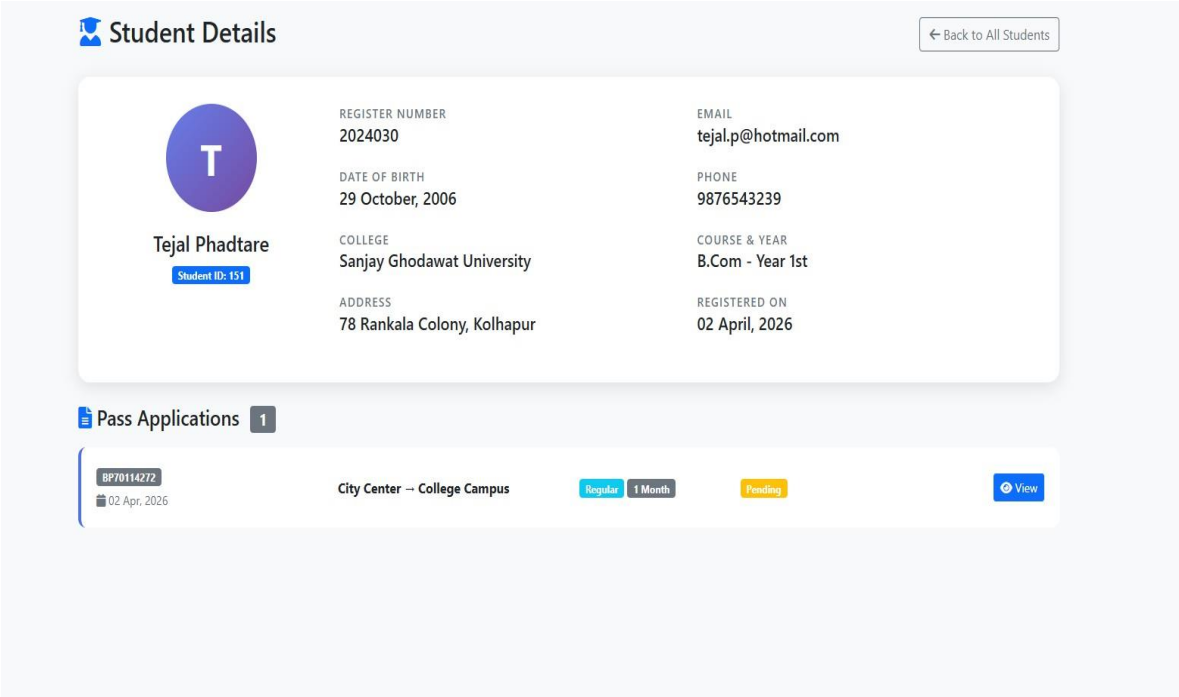
7.9 All Students (Registered on portal)



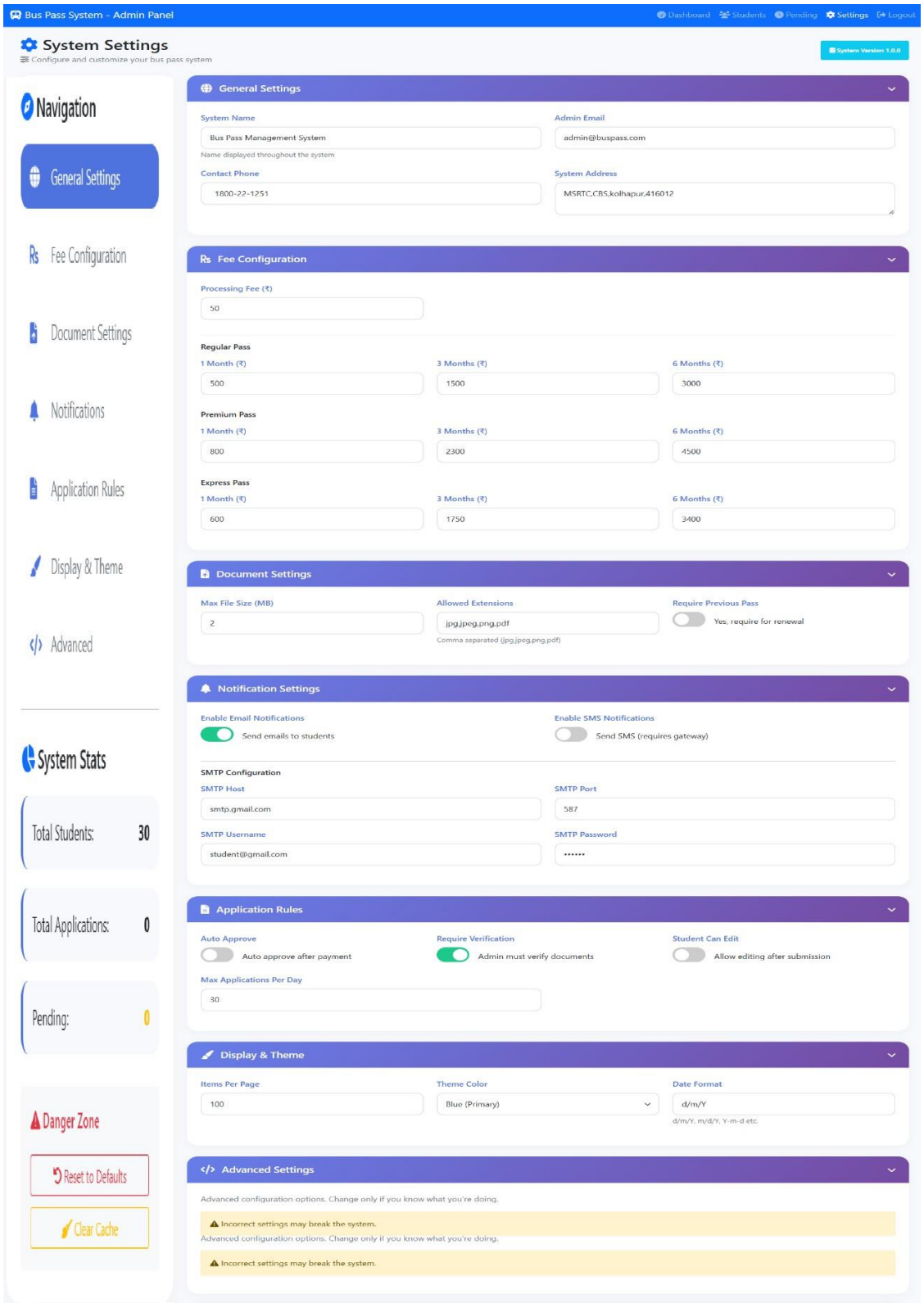
7.10 Applications Pending



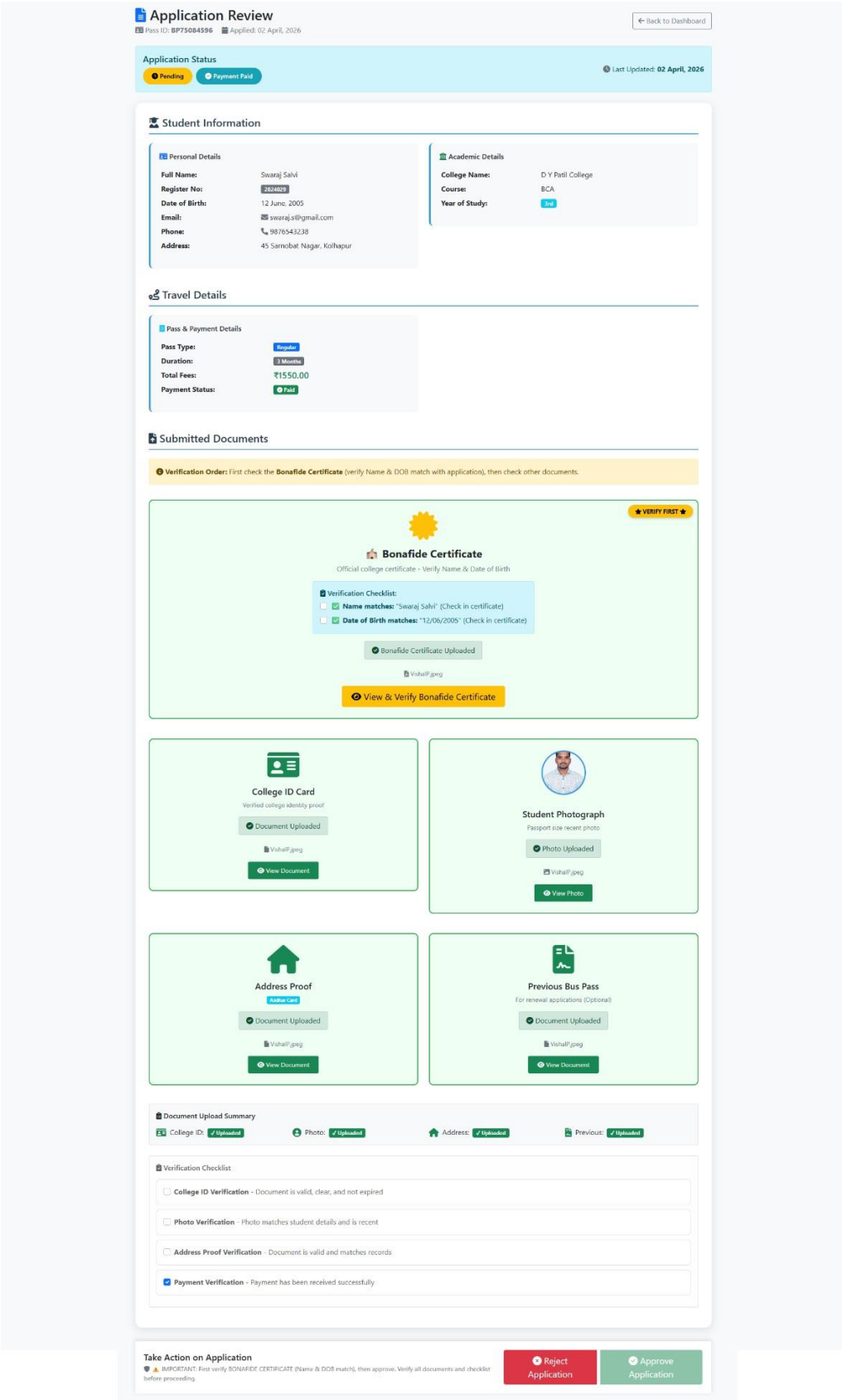
7.11 Student Details with their Pass History



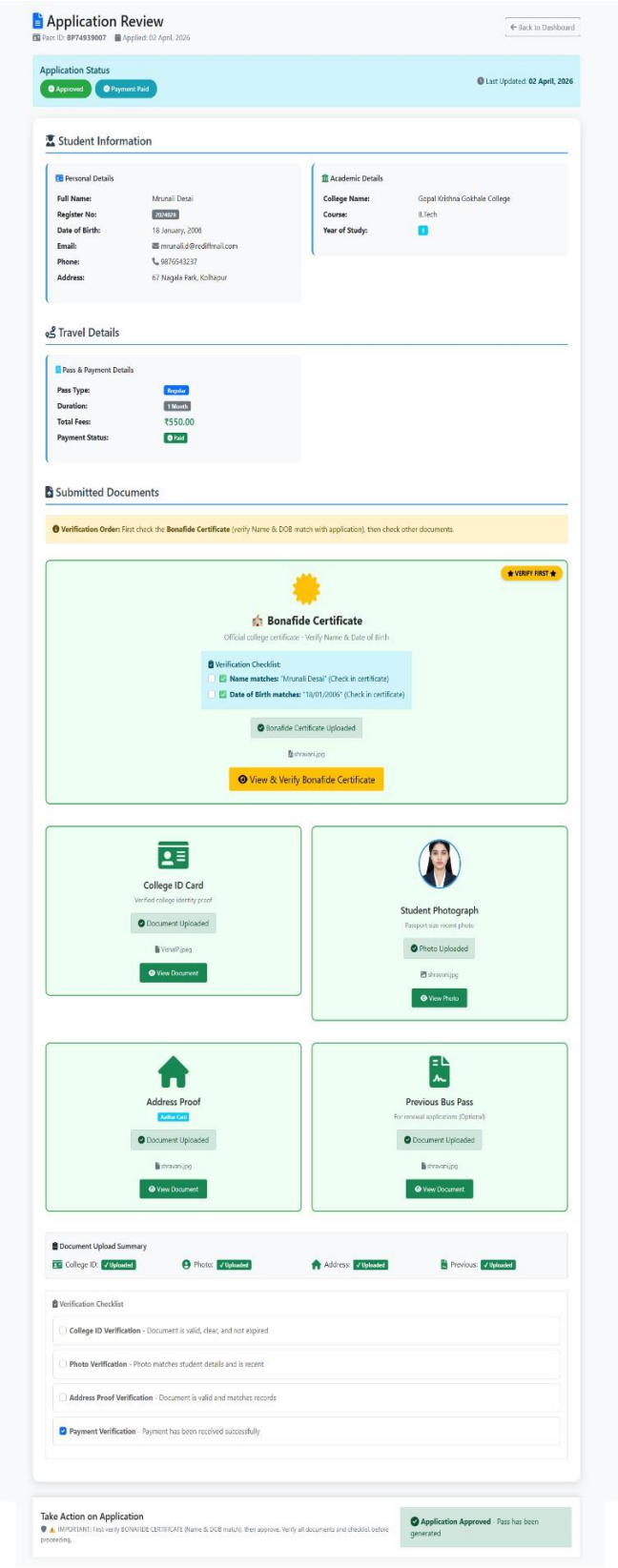
7.12 Settings Of Portal (Admin can handle portal)



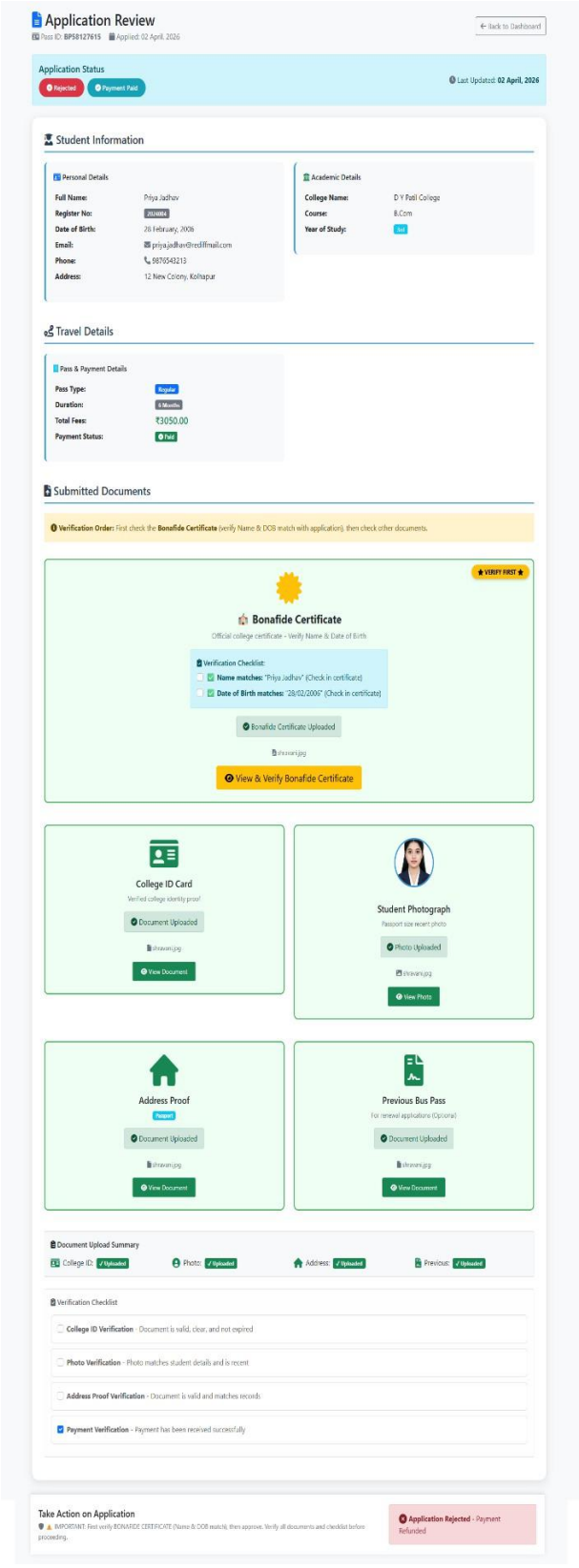
7.13 Application Review For Verify



Approved Application



Rejected Application



7.14 The Bus Pass

MAHARASHTRA STATE ROAD TRANSPORT CORPORATION

Gopal Krishna Gokhale

BUS PASS • ACADEMIC YEAR 2025-26

Regular

Student Name	Vishal Potdar
Register No.	5845
Course & Year	BCA - Year 1
Date of Birth	14/05/2002
Route	Rendal → Shahu Maidan
Pass Duration	1 Month
Pass ID	BP84676528

Scan to verify

CONDUCTOR VERIFICATION

Unique Pass ID: BP84676528

Scan QR code or visit collegebuspass.com/verify and enter Pass ID.

GOPAL KRISHNA GOKHALE COLLEGE, KOLHAPUR APPROVAL KOLHAPUR

College Authority

Validity: 03/04/2026 - 03/05/2026

Electronically generated, no signature required.

Chapter 8:
Conclusion & Suggestions

8.1 CONCLUSION

The Maharashtra Bus Pass System represents a significant milestone in the digital transformation of public transportation services within the state. This project successfully addresses the longstanding challenges faced by students, educational institutions, and the Maharashtra State Road Transport Corporation in the manual bus pass issuance process. Through the development of a comprehensive web-based platform, the system has effectively eliminated the inefficiencies of paperwork, reduced processing times from weeks to days, and introduced unprecedented transparency in application tracking and approval workflows.

Key Contributions

- √ **Simplified student registration and application process**
- √ **Streamlined institution verification workflow**
- √ **Automated transport authority approval system**
- √ **Secure digital pass generation with unique QR codes**
- √ **Mobile-friendly validation for bus conductors**
- √ **Comprehensive admin dashboard for monitoring and reporting**
- √ **Integrated payment processing with receipt generation**
- √ **Configurable fare slabs and system settings**
- √ **Audit logging for accountability and security**
- √ **Responsive design accessible across all devices**

In conclusion, the Maharashtra Bus Pass System stands as a testament to how technology can transform public services, making them more accessible, efficient, and secure. The project has successfully delivered a solution that not only meets the immediate needs of students and transport authorities but also lays a strong foundation for future innovations in public transportation management.

The system is poised to make a lasting positive impact on the lives of millions of students across Maharashtra. The successful completion of this project demonstrates the power of thoughtful software engineering in solving real-world problems and serves as a model for similar initiatives in other states and sectors.

8.2 FUTURE ENHANCEMENT

The Maharashtra Bus Pass System can be further improved by adding advanced features and expanding its functionality.

The following enhancements can be implemented in the future:

1. **Mobile Application Development**
   The system can be converted into a mobile application for easier access and better user experience.
2. **Integration with Government Systems**
   The system can be connected with official government databases for better verification and authenticity.
3. **Multi-State Support**
   The system can be expanded to other states by integrating with their respective transport departments.
4. **Advanced Security Features**
   Additional security features such as OTP verification and data encryption can be implemented.
5. **Admin Analytics Dashboard**
   A dashboard can be added to provide reports and statistics for better decision-making by authorities.
6. **Digital Document Verification**
   The system can include a feature to upload and verify student documents such as ID cards and certificates.
7. **Automatic Pass Expiry Management**
   The system can automatically track pass validity and notify users before expiry.
8. **Cloud-Based Deployment**
   The system can be hosted on cloud platforms to improve accessibility, performance, and scalability.
9. **Route and Bus Tracking System**
   Real-time tracking of buses and routes can be added to improve convenience for users.

Overall, these future enhancements will make the system more efficient, scalable, and suitable for real-world government implementation.

8.3 BIBLIOGRAPHY

The following sources were referred to during the development of the Maharashtra Bus Pass System:

1. Python Flask Official Documentation
   https://flask.palletsprojects.com/

2. W3Schools – HTML, CSS, and SQL Tutorials
   https://www.w3schools.com/

3. GeeksforGeeks – Programming and Database Concepts
   https://www.geeksforgeeks.org/

4. Stack Overflow – Technical Discussions and Problem Solving
   https://stackoverflow.com/

5. MDN Web Docs – HTML, CSS, and Web Standards
   https://developer.mozilla.org/

6. Tutorialspoint – Web Development and Database Tutorials
   https://www.tutorialspoint.com/

7. AI Assistance Tools (such as ChatGPT) for conceptual understanding and content guidance

8. YouTube Tutorials for Flask and Web Development Concepts

9. Class Notes and Study Materials provided by Faculty

These resources helped in understanding the concepts of web development, system design, and database management required for the successful completion of this project.